When Electric Vehicle Off-Peak Charging Discount Backfire and When They Don't: Experimental Evidence from South Korea*

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Abstract

We conduct a randomized controlled trial with 453 electric vehicle drivers on a commercial charging platform in South Korea, randomly assigning participants to off-peak charging discounts (22:00–08:00) that vary in intensity (20% or 40%) and duration (one or two months), or to a no-incentive control group. A 40% discount sustained for two months raises the daily probability of off-peak charging by 3 percentage points—a 150% increase relative to the control mean—and increases off-peak charging minutes by 196%. A 20% discount produces small and largely insignificant short-run effects, pointing to a nonlinear incentive–response relationship in which a sufficient price signal is needed to overcome behavioral inertia. The effects concentrate on slow chargers, consistent with overnight home charging as the primary behavioral margin, and we find no consistent reduction in non-off-peak charging on the platform. Because our data capture only sessions conducted through the partner platform, we interpret these estimates as effects on platform-observed off-peak charging rather than on aggregate system load. Our results suggest that residential slow charging is a scalable behavioral margin for demand response in dense urban markets, whereas the environmental sign of such incentives in coal-backed overnight grids ultimately depends on the carbon intensity of the marginal generator and warrants careful design of the off-peak window.

JEL Codes: Q41, Q42, Q48, R41

Keywords: electric vehicles, field experiment, off-peak charging, dynamic electricity pricing, demand response

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1 Introduction

Electric vehicle (EV) adoption is accelerating worldwide. Global EV sales exceeded 17 million units in 2024, and the fleet is projected to continue growing as governments pursue decarbonization of the transportation sector (International Energy Agency 2024; Jaramillo et al. 2022). While this transition promises substantial reductions in tailpipe emissions, it introduces a less visible challenge for electricity systems. The concern is not simply that EVs raise total electricity demand but also that charging load concentrates in time. When large numbers of drivers plug in during the same hours, the resulting demand spike strains generation capacity, distribution infrastructure, and grid stability (International Energy Agency 2022; U.S. Department of Energy, Office of Electricity 2024). A defining feature of electricity markets is that supply and demand must balance instantaneously; unlike most commodities, electricity cannot be economically stored at scale, making the temporal distribution of load as consequential as its level (Borenstein 2002). The "duck curve" phenomenon—where midday solar surplus is followed by a sharp evening ramp in net demand—compounds this problem, as EV charging that clusters in the early evening deepens the trough-to-peak swing (California Independent System Operator 2016).

Spreading EV charging across hours, and particularly shifting it to off-peak periods, is therefore a first-order policy objective. Among the instruments available (i.e., time-of-use (TOU) tariffs, automated managed charging, informational nudges), financial incentives tied to off-peak windows are the most straightforward to implement within existing billing infrastructure. Yet evidence on whether such incentives work, how large they must be, and whether their effects persist remains limited. We address these questions through a randomized controlled trial (RCT) conducted in South Korea, a setting with several features that make it a compelling test case. South Korea's EV fleet exceeded 543,900 vehicles and 290,000 chargers by the end of 2023, with high urban density and predominantly apartment-based slow charging creating a distinctive infrastructure landscape (Woo and Y. J. Kim 2025). The country operates an explicit TOU tariff for EV charging, yet a substantial share of charging continues outside the off-peak window (22:00–08:00), suggesting that the existing rate differential is insufficient to overcome habitual behavior.

In our experiment, 453 EV drivers on Technology Beyond Utility's (TBU) Elecvery platform, a commercial roaming EV charging service with approximately 600,000 monthly active users that aggregates real-time information on over 520,000 chargers nationwide, were randomly assigned to one of four treatment arms or a no-incentive control group. The treatment arms varied along two dimensions: discount intensity (20% versus 40% off the standard charging rate) and duration (one month versus two months). The discounts applied exclusively to charging sessions initiated during off-peak hours (22:00–08:00). For an eligible session, the discount applied to the entirety of the energy delivered by that session, regardless of whether the session subsequently extended into peak hours. This implementation detail is platform-driven and is important context for interpreting the magnitude of effects measured in energy and expenditure outcomes. We observe detailed charger-level log data (i.e., time-stamped records of every charging session, including energy delivered (kWh), duration, and charger type (fast or slow)), spanning both pre- and post-treatment periods.

One important caveat is that our data capture only charging sessions conducted through the TBU platform; we cannot observe whether participants also charge through other operators. This means that measured increases in off-peak charging on TBU could in principle reflect substitution across platforms rather than a net change in charging behavior. We discuss this limitation throughout and treat the empirical estimand as the change in platform-observed off-peak charging rather than the change in total system load.

We find three main results. First, a 40% discount sustained for two months (T4) generates large and statistically significant increases in off-peak charging on the platform. During the first month post-treatment, T4 recipients are 3 percentage points more likely to charge during off-peak hours on a given day, a 150% increase relative to the control-group mean of 0.02, and they charge 7.14 additional off-peak minutes per day, a 196% increase. The effects persist and grow through the second month: the daily off-peak minutes coefficient rises to 9.58, equivalent to a 262% increase over the control mean. Second, a 40% discount applied for only one month (T2) produces moderate effects that are statistically significant for some outcomes, whereas a 20% discount produces small and largely insignificant short-run effects; only when the 20% arm is sustained for two months (T3) does it generate marginally significant effects on energy and expenditure outcomes. This pattern suggests a nonlinear incentive–response

relationship in which a 20% reduction is insufficient to overcome behavioral inertia in the short run, whereas a 40% reduction crosses a threshold that induces meaningful change quickly; a smaller discount can partially compensate only when exposure is long enough to allow new charging routines to form. Third, we find no consistent reduction in non-off-peak charging on the platform, which suggests that the response is closer to additive than substitutive within TBU. Because our data capture only charging sessions conducted through the platform, we cannot rule out that part of the measured off-peak increase as reflecting a substitution effect with competing operators, thus we interpret our estimates as effects on platform-observed off-peak charging rather than on aggregate system load.

Our paper contributes to a growing experimental literature on electricity demand management. A large body of work examines time-varying pricing for residential electricity (Jones et al. 2022) and grid management (Li and Jenn 2024). Faruqui and Sergici (2010) survey 15 dynamic pricing pilots and find peak-demand reductions of 3–6% under TOU rates and 13–20% under critical-peak pricing; Faruqui and Sergici (2017) extend this analysis to 63 pilots across nine countries, confirming that the peak-to-off-peak price ratio is the key predictor of demand response. Ito (2014) demonstrates that consumers respond to average rather than marginal electricity prices under nonlinear tariff schedules, a finding with direct implications for the design of EV charging incentives. Jessoe and Rapson (2014) show experimentally that real-time usage information amplifies price elasticities, suggesting that salience matters alongside price levels. Hofmann and Lindberg (2024) provide recent evidence from a large field experiment in Norway, documenting household demand flexibility under variable hourly pricing.

A nascent strand of this literature focuses specifically on EV charging behavior (Burkhardt et al. 2019). Bailey et al. (2025b) conduct a field experiment in Calgary, Canada, finding that a financial incentive of approximately 23% reduces peak-hour charging by 49%, whereas a prosocial information treatment has no discernible effect; critically, charging behavior reverts to baseline once incentives are removed. Bailey et al. (2025a) demonstrate that TOU pricing, while effective at shifting EV charging into off-peak windows, creates unintended "shadow peaks" of simultaneous charging that strain local distribution networks. Bernard et al. (2025) study dynamic pricing for public EV charging across 110,000 customers in the United King-

dom, finding that a 40% cost reduction increases charging activity by 117%. Metcalfe et al. (2026) evaluate an AI-managed charging system for over 13,000 UK consumers, showing that automated load management shifts 100% of charging to low-cost periods and reduces peak demand by 42%. A particularly informative counterpoint is found with Garg et al. (2025), who conducted a workplace charging experiment with 629 drivers at University of California, San Diego and find that flat (time-invariant) financial discounts shift charging away from the solar peak, raising early-morning, evening, and overnight sessions and increasing CO_2 emissions per driver under the prolonged discount, while an environmental nudge re-times charging modestly toward midday. The setting differs from ours along three dimensions simultaneously: instrument design (flat versus time-conditional), dominant charging margin (workplace daytime versus residential overnight), and grid composition (solar-abundant California versus fossil-backed overnight Korea). Reading the two experiments side by side suggests that the environmental sign of a charging incentive is jointly determined by these features rather than by any one of them, and that residential time-conditional discounts can mobilize a behavioral margin (i.e., overnight slow charging at home) that flat workplace discounts cannot reach. We return to this comparison in Section 4.

Although a growing experimental literature has begun to estimate the price elasticity of EV charging demand in North American (Bailey et al. 2025a; Bailey et al. 2025b; Garg et al. 2025) and European (Bernard et al. 2025; Metcalfe et al. 2026) settings, comparable evidence from still-novel environments such as dense, apartment-based Asian markets remain limited. We extend this literature in three ways.

First, we provide the first experimental evidence on EV charging incentives in an Asian economy. South Korea's charging landscape differs from previously studied settings along three dimensions that jointly limit the external validity of existing estimates: high urban density, predominantly apartment-based slow charging (approximately 90% of Korean chargers are slow chargers as of mid-2026), and a time-of-use (TOU) tariff schedule in which an off-peak window of 22:00–08:00 is already priced separately from peak and mid-peak hours. Whether financial incentives can shift behavior on top of this existing rate differential is, therefore, an open question with direct policy relevance for dense urban markets across East and Southeast Asia. Second, by experimentally varying both discount intensity (20% versus

40%) and duration (one versus two months) within a single factorial design, we identify a nonlinear incentive–response relationship: a 20% discount produces little measurable change in the short run, whereas a 40% discount crosses a threshold that induces a substantial behavioral shift. This pattern suggests that marginal adjustments to existing TOU rate differentials are unlikely to move charging behavior, and that policy makers seeking demand response should consider the magnitude of the price signal alongside its temporal structure. Third, granular charger-level data allow us to decompose treatment effects by charger type (i.e., speed, or power output levels), and we find that the response concentrates on slow chargers used primarily for overnight home charging. Because slow chargers account for the vast majority of the Korean network and require only a modest adjustment in plug-in timing (i.e., waiting until 22:00 rather than plugging in upon arriving home in the early evening), the relevant behavioral margin is technologically straightforward and broadly scalable, with implications for the persistence and policy reach of such interventions as the EV fleet continues to grow.

The paper proceeds as follows. Section 2 describes the study setting, experimental design, and data. Section 3 presents the results, including main treatment effects, heterogeneity analyses, and event-study evidence. Section 4 discusses policy implications. Section 5 concludes.

2 Study Design and Data

2.1 Setting

The transportation sector is a major contributor to climate change, accounting for approximately 15% of global greenhouse gas emissions (Jaramillo et al. 2022). In South Korea, the sector represents roughly 20% of national energy-related CO_2 emissions. The transition toward EVs has emerged as a cornerstone of domestic decarbonization strategies, and the government has supported adoption through purchase subsidies and infrastructure investment (Y. J. Kim et al. 2026). South Korea's EV market has experienced rapid expansion, with the total fleet exceeding 543,900 vehicles and the supporting infrastructure growing

to nearly 290,000 chargers by the end of 2023 (Woo and Y. J. Kim 2025). This growth momentum has continued, with EVs representing a record 3.53% of all vehicle registrations as of the end of February 2026 (Ministry of Land, Infrastructure and Transport 2026).

The infrastructure landscape is shaped by high urban density and concentrated residential patterns, particularly in apartment complexes where the majority of the population resides. Slow chargers constitute the vast majority of the network. Residential slow chargers are primarily used for overnight charging, while non-residential fast chargers see peak usage during daytime hours. This distribution creates a distinct charging profile compared to other developed countries and necessitates localized strategies for grid management.

Table 1 presents the electricity tariff structure for EV charging in 2025. The time-of-use schedule defines off-peak hours as 22:00–08:00, with mid-peak and on-peak hours during daytime and early evening. The pricing structure is designed to encourage charging during off-peak periods in order to reduce peak load pressures on the grid. Despite these price differentials, a substantial fraction of EV charging continues to occur outside of off-peak hours, suggesting that the existing rate differential may be insufficient to overcome habitual charging patterns. For context, Korea's residential electricity pricing uses a three-tier progressive tariff system, where electricity becomes significantly more expensive as consumption exceeds 300 kWh and 450 kWh per month. Compared with the United States and European Union, Korea generally has lower average electricity prices but a steeper progressive structure.

Table 1: Customer for charging electric vehicles

Classification	Demand charge (KRW/kW)	Time period	Energy charge (KRW/kWh)		
			Summer	Spring/Fall	Winter
Low-voltage	2,390	off-peak	76.3	77.4	99.4
		mid-peak	164.0	89.2	146.9
		on-peak	251.2	94.1	209.5
High-voltage	2,580	off-peak	71.2	72.2	88.6
		mid-peak	129.4	83.0	119.7
		on-peak	182.4	86.9	157.5

Note: Time-of-use periods are defined by season and hour of day. Summer includes June–August; spring includes March–May; fall includes September–October; and winter includes November–February. Off-peak hours are 22:00–08:00 in all seasons. In summer/spring/fall, mid-peak hours are 08:00–11:00, 12:00–13:00, and 18:00–22:00, while on-peak hours are 11:00–12:00 and 13:00–18:00. In winter, mid-peak hours are 08:00–09:00, 12:00–16:00, and 19:00–22:00, and on-peak hours are 09:00–12:00 and 16:00–19:00.

A critical distinction for demand management policy is between fast chargers and slow chargers. Fast chargers, typically located at public charging stations, are generally able to deliver a full charge in 30 to 60 minutes and are often used opportunistically during daytime travel, and for individuals who do not have access to "home" charging. Slow chargers, typically installed at apartment complexes and other residential facilities, charge overnight over the course of several hours. This asymmetry means that policies targeting slow home chargers may yield larger behavioral responses on the time-shifting margin than those targeting fast public chargers, since overnight charging requires only a modest adjustment in plug-in timing.

2.2 Intervention

This experiment is conducted in partnership with TBU, Inc., an EV charging platform that provides charging station information and roaming charging services in Korea. In late July 2025, an eligible study population was identified from TBU's membership database, with eligibility criteria explained in the subsequent section. The intervention itself consisted of a subsidy of either 20 or 40% of the standard charging rate, in the form of a platform-usable cashback of equivalent value, applied at the platform level whenever a participant

initiated a charging session within the 22:00-08:00 window at a charger operated through the TBU network. The discount was applied to the full session bill, i.e., to all energy delivered and billed during the session, irrespective of whether the session continued past 08:00. This session-initiation rule reflects how the discount was implemented in TBU's billing infrastructure and matches what participants saw at the point of sale. Treatment groups were informed via direct messaging (i.e., text messaging and KakaoTalk notifications), and participants did not need to enter codes or take additional steps beyond plugging in during the eligible window.

2.3 Experimental Design

The study population was drawn from the approximately 600,000 monthly active users (MAU) of the Elecvery platform. Eligibility required (i) at least three months of EV ownership, (ii) at least KRW 5,000 of recorded charging activities with the Elecvery service within the past 30 days from the time of survey completion, (iii) consent to share certain personally identifying information (name, mobile number, primary account type), and (iv) the full completion of the baseline survey.

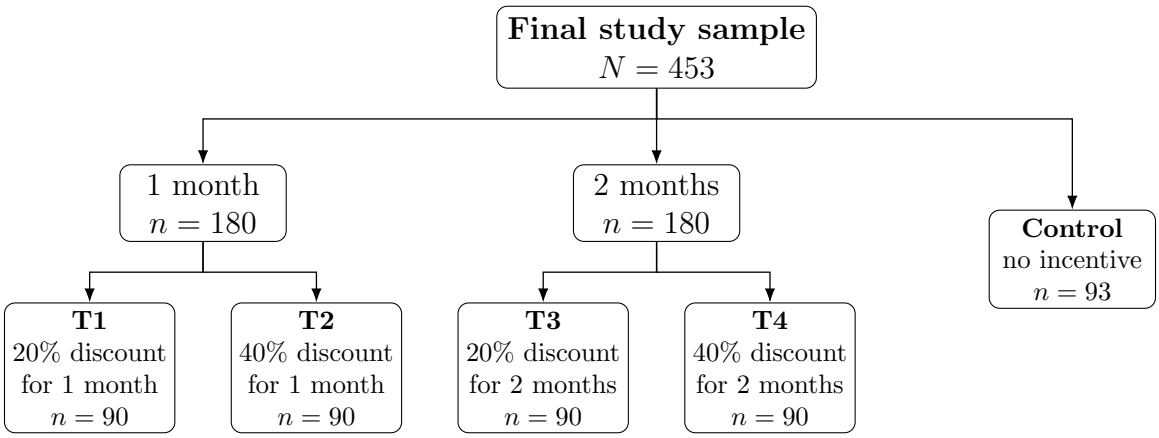
The baseline survey was administered between August 1 and August 17, 2025. The first month of the treatment period began at the close of the survey window and runs through August-September 2025; the second month continues through October 2025 for the two-month arms (T3 and T4). Post-treatment follow-up data collection continues through December 2025.

Following baseline survey completion, 453 participants were randomly assigned to one of five conditions: four treatment arms (T1–T4) and a no-incentive control group. The treatment arms vary along two dimensions: discount intensity (20% or 40%) and duration (one month or two months). Specifically, T1 offers a 20% discount for one month ($n = 90$), T2 offers a 40% discount for one month ($n = 90$), T3 offers a 20% discount for two months ($n = 90$), and T4 offers a 40% discount for two months ($n = 90$). The control group ($n = 93$) receives no incentive. Figure 1 summarizes the design.

The experiment proceeded in two phases following a pre-treatment observation period spanning May–July 2025. The first month of treatment ran through August 2025; the

second month continued through September 2025 for the two-month arms (T3 and T4). Post-treatment follow-up data collection continued through December 2025. This structured design allows us to separately identify the effects of incentive intensity and duration, and to examine whether behavioral changes persist after incentives are removed.

Figure 1: Experimental design and sample sizes



2.4 Data

We draw on two primary data sources. First, a baseline survey was administered August 1-17, 2025, immediately prior to treatment assignment. The survey collected information on demographics (age, gender, education, region of residence and workplace), vehicle characteristics (EV model, years of ownership, annual mileage), infrastructure access (availability of fast and slow chargers at home and at work), and self-reported charging habits (most frequent charging location and time of day, monthly charging frequency and expenditure).

Second, we use detailed charger log data provided, provided by TBU Inc., spanning May 2024 to December 2025, covering both pre-treatment and post-treatment periods. These administrative records capture every charging session on the TBU platform at the user level, including start and end timestamps, total energy delivered (kWh), charger type (fast or slow), and charged amount. The data enable precise measurement of the timing, duration, and energy content of each charging event. Our primary unit of analysis is the user-day: for each participant and calendar day, we construct outcome variables summarizing off-peak (22:00–08:00) and non-off-peak charging activity.

Session-day construction. Our primary unit of analysis is the user-day. For each participant and calendar day, we construct outcome variables summarizing off-peak (22:00–08:00) and non-off-peak charging activity. Because charging sessions can cross the 22:00, midnight, or 08:00 boundaries, we adopt the following coding rule. Each session is split into within-window and outside-window minutes and energy based on the actual clock-time overlap; these split components are then attributed to the calendar day on which the corresponding minutes/energy occurred. For example, a session from 21:30 to 02:30 contributes 30 minutes (and the corresponding pro-rata energy and cost) to the non-off-peak block of day t, 120 minutes to the off-peak block of day t, and 150 minutes to the off-peak block of day $t+1$. This clock-time apportionment is independent of the discount-eligibility rule (which is session-initiation based) and is applied uniformly to treatment and control. Robustness checks using a session-initiation rule for outcome construction (i.e., assigning the entire session to whichever block contains the start time) yield qualitatively similar estimates.

2.5 Baseline Summary Statistics and Balance

Table 2 assesses whether treatment groups are balanced relative to the control across baseline characteristics. Panel A shows no statistically significant differences in gender, age distribution, education, or residential and workplace location. Panel B confirms balance on EV-related characteristics, including ownership duration and annual mileage. Panel C reports charger access: roughly 57% of the control group has access to a fast charger at home and 84% to a slow charger, with no significant differences across arms. Panel D examines self-reported charging behavior—most frequent charging location, monthly frequency, expenditure, and time of day—and again finds no significant imbalances. The randomization appears successful in producing comparable groups across all observable dimensions.

Table 2: Balance Checks across Samples

	(1) Mean (SD)	(2)	(3)	(4)	(5)
		Coefficient on Difference from Control (s.e.)			
	Control group	T1 (20%, 1mo)	T2 (40%, 1mo)	T3 (20%, 2mo)	T4 (40%, 2mo)
Panel A. Respondent demographics					
Female	0.17	-0.03	-0.04	-0.07	-0.05
		(0.05)	(0.05)	(0.05)	(0.05)
Age: 20-39	0.32	-0.03	0.03	-0.06	-0.01
		(0.07)	(0.07)	(0.07)	(0.07)
Age: 40-59	0.60	0.04	0.02	0.09	-0.01
		(0.07)	(0.07)	(0.07)	(0.07)
Age: 60 or above	0.08	-0.01	-0.04	-0.03	0.02
		(0.04)	(0.03)	(0.04)	(0.04)
College or above	0.90	-0.09	-0.04	-0.04	-0.03
		(0.05)	(0.05)	(0.05)	(0.05)
Resides in Seoul Metro Area	0.54	-0.04	-0.01	-0.02	0.01
		(0.07)	(0.07)	(0.07)	(0.07)
Works in Seoul Metro Area	0.51	-0.02	0.02	0.02	0.04
		(0.07)	(0.07)	(0.07)	(0.07)
Resides in APT	0.68	0.10	-0.00	0.06	0.04
		(0.07)	(0.07)	(0.07)	(0.07)
Panel B. EV characteristics					
EV own:					
<1 year	0.12	0.00	-0.05	-0.03	-0.01
		(0.05)	(0.04)	(0.05)	(0.05)
1-3 years	0.59	-0.08	-0.04	-0.11	-0.10
		(0.07)	(0.07)	(0.07)	(0.07)
3+ years	0.14	0.08	0.10	0.07	0.09
		(0.06)	(0.06)	(0.06)	(0.06)
Annual mileage:					
<10,000 km	0.23	0.05	0.12	0.03	0.06
		(0.06)	(0.07)	(0.06)	(0.06)
10,000-15,000 km	0.16	-0.02	0.07	0.14	0.08
		(0.05)	(0.06)	(0.06)	(0.06)
15,000+ km	0.61	-0.04	-0.20	-0.17	-0.15
		(0.07)	(0.07)	(0.07)	(0.07)
Panel C. EV charger access					
At home:					
Fast charger	0.57	-0.04	-0.08	-0.11	-0.08
		(0.07)	(0.07)	(0.07)	(0.07)
Slow charger	0.84	0.04	0.00	0.02	-0.05
		(0.05)	(0.05)	(0.05)	(0.06)
At work:					
Fast charger	0.45	-0.02	-0.05	-0.14	-0.03
		(0.07)	(0.07)	(0.07)	(0.07)
Slow charger	0.60	-0.00	0.07	-0.00	0.05
		(0.07)	(0.07)	(0.07)	(0.07)
Panel D. EV charging behavior					
Most frequent charge location:					
Home	0.42	0.08	0.10	0.10	0.04
		(0.07)	(0.07)	(0.07)	(0.07)
Work	0.22	0.02	-0.01	0.04	-0.03
		(0.06)	(0.06)	(0.06)	(0.06)
Public spaces	0.33	-0.09	-0.09	-0.14	-0.01
		(0.07)	(0.07)	(0.06)	(0.07)
Number of charging per month:					
1-5 times	0.37	0.08	0.16	-0.02	0.00
		(0.07)	(0.07)	(0.07)	(0.07)
6-9 times	0.29	-0.06	-0.08	0.04	0.04
		(0.07)	(0.06)	(0.07)	(0.07)
10+ times	0.34	-0.02	-0.09	-0.02	-0.04
		(0.07)	(0.07)	(0.07)	(0.07)
Monthly charge expenses:					
<50,000 KRW	0.35	-0.02	0.07	0.09	0.00
		(0.07)	(0.07)	(0.07)	(0.07)
50,000-100,000 KRW	0.41	-0.03	-0.06	-0.01	0.05
		(0.07)	(0.07)	(0.07)	(0.07)
100,000+ KRW	0.48	-0.06	-0.15	-0.16	-0.08
		(0.07)	(0.07)	(0.07)	(0.07)
Most frequent charge time:					
06:00-12:00	0.16	-0.01	-0.02	-0.01	-0.01
		(0.05)	(0.05)	(0.05)	(0.05)
12:00-18:00	0.23	0.02	0.02	0.02	0.02
		(0.06)	(0.06)	(0.06)	(0.06)
18:00-22:00	0.61	-0.01	-0.02	-0.01	-0.01
		(0.07)	(0.07)	(0.07)	(0.07)
Observations	93	90	90	90	90

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2.6 Empirical Strategy

We estimate the following two-way fixed effects (TWFE) specification:

$$Y_{it} = \sum_{k=1}^{4} \beta_k \left(\text{Incentive}_{k,i} \times \text{Post}_t \right) + \delta_i + \tau_t + \varepsilon_{it} \tag{1}$$

where Y_{it} is the outcome for user i on day (or week) t; $\text{Incentive}_{k,i}$ is an indicator for assignment to treatment arm $k \in \{T1, T2, T3, T4\}$; Post_t equals one for dates in the post-treatment period; δ_i denotes individual fixed effects; and τ_t denotes time (day or week) fixed effects. The coefficients of interest are β_k, which capture the intention-to-treat effects of each incentive arm relative to the control group. Standard errors are clustered at the individual level.

Our primary outcomes are: (1) an indicator equal to one if user i engages in any off-peak charging (22:00–08:00) on day t; (2) total minutes charged during off-peak hours; (3) total energy charged off-peak (kWh); and (4) total expenditure on off-peak charging (KRW). We also examine corresponding non-off-peak outcomes (all hours outside 22:00–08:00) and total charging, which allow us to test whether incentivized off-peak charging comes at the expense of reduced charging at other times (substitution) or represents a net increase (additive response). In addition to the pooled TWFE estimates, we present event-study specifications that replace Post_t with a full set of weekly indicators relative to the treatment start date, allowing us to trace the dynamic evolution of treatment effects and assess pre-treatment parallel trends.

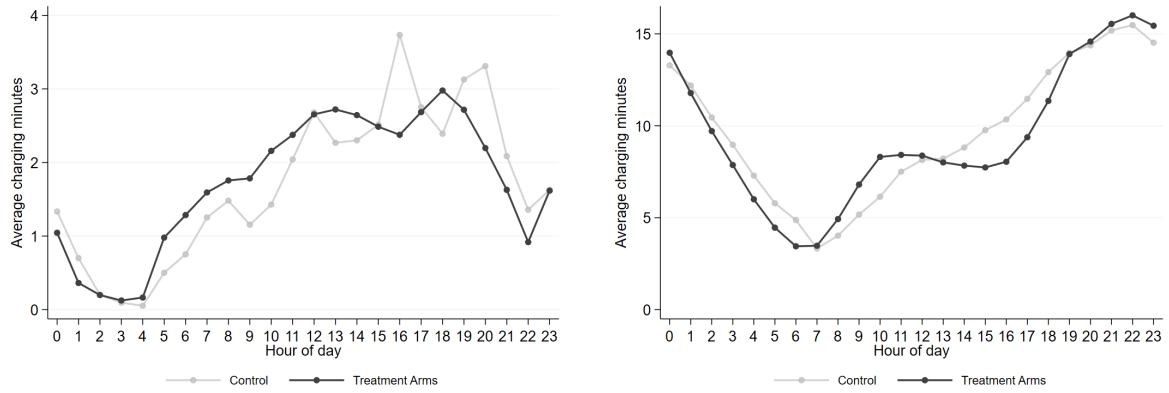
3 Results

3.1 Descriptive Evidence

Before turning to regression estimates, we present descriptive evidence on charging patterns. Figure 2 plots the average pre-treatment hourly distribution of charging minutes for treatment (combined) and control groups, separately for fast and slow chargers. The distributions are visually similar across groups, consistent with the balance results in Table 2 and

supporting the parallel-trends assumption underlying our identification strategy.

Figure 2: Average Pre-Treatment Charging Minutes: Treatment Arms Combined vs. Control



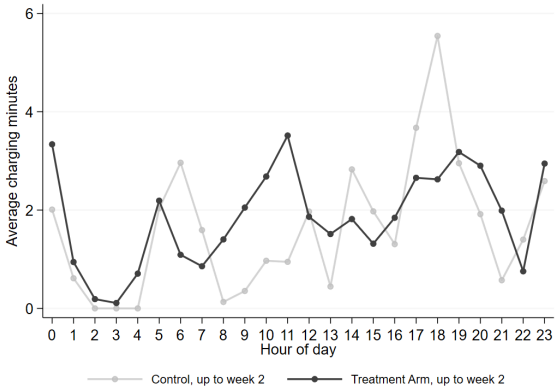
(a) Fast chargers

(b) Slow chargers

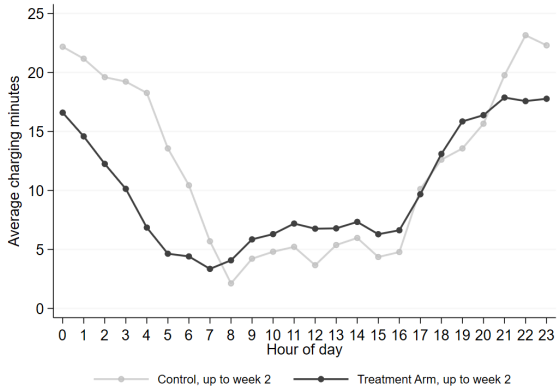
Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group (all arms combined).

Figure 3 displays the same hourly distributions for the post-treatment period, at two- and four-week windows. Treatment-group drivers show a visible shift toward late-night and early-morning charging on both fast and slow chargers, particularly at the four-week horizon. The shift is most pronounced on slow chargers, where treatment-group charging during the 22:00–08:00 window increases noticeably relative to the control.

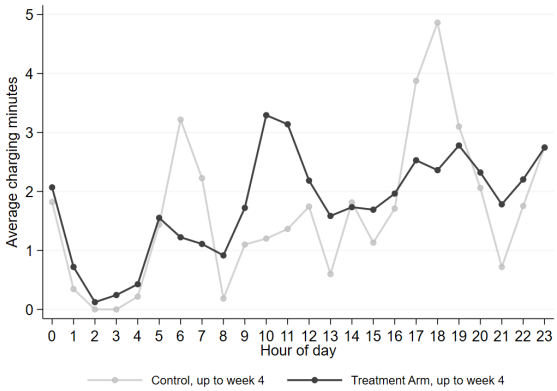
Figure 3: Average Post-Treatment Charging Minutes: Treatment Arms Combined vs. Control



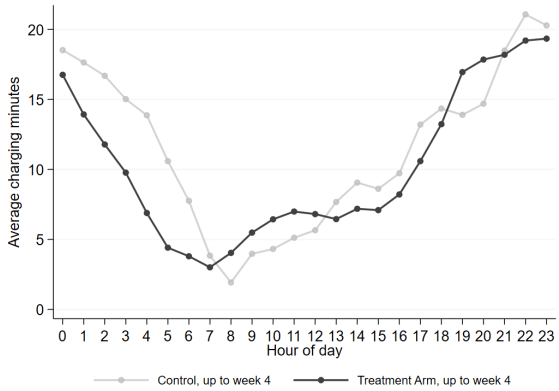
(a) Fast chargers (up to 2 weeks post-treatment)



(b) Slow chargers (up to 2 weeks post-treatment)



(c) Fast chargers (up to 4 weeks post-treatment)

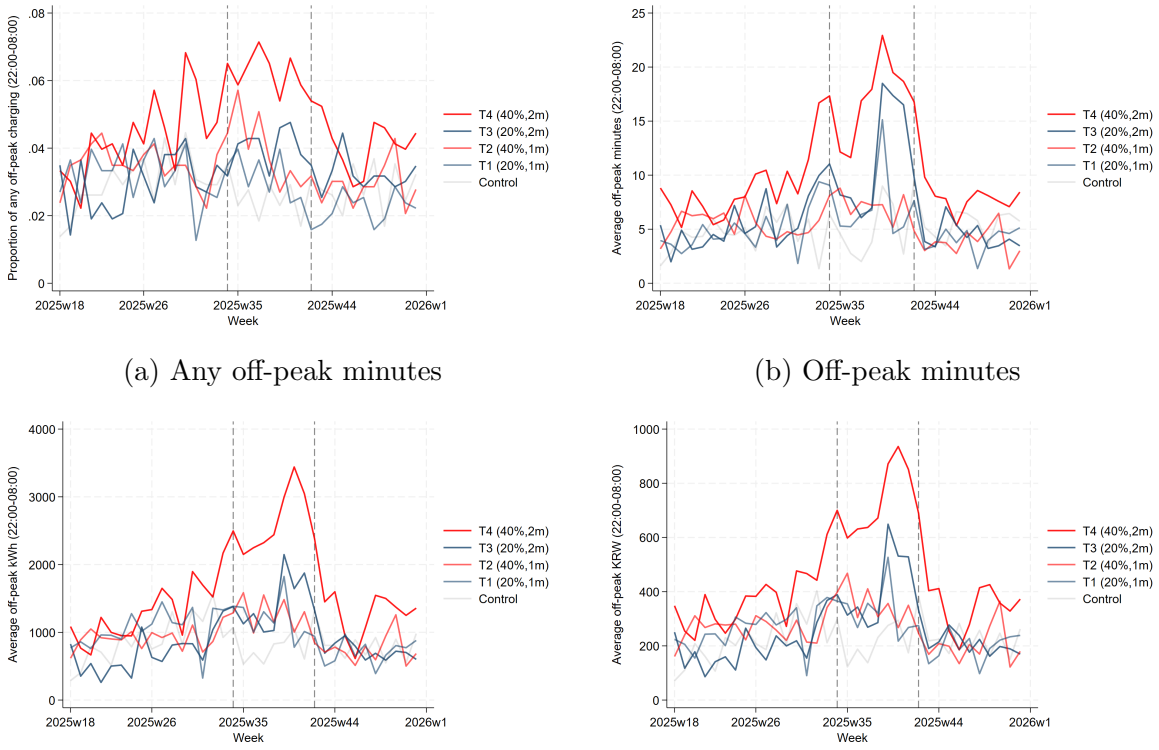


(d) Slow chargers (up to 4 weeks post-treatment)

Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group (all arms combined).

Figure 4 presents weekly time series of off-peak outcomes (e.g., any off-peak charging, total minutes, kWh, and expenditure in Korean Won) for the (combined) treatment and control groups. The series track each other closely in the pre-treatment period and diverge after treatment begins, with the treatment group exhibiting higher off-peak charging across all four measures.

Figure 4: Weekly Off-peak Outcomes (22:00–08:00)



(a) Any off-peak minutes

(b) Off-peak minutes

(c) Off-peak kWh

(d) Off-peak KRW

Note: Each panel plots weekly averages for treatment (all arms combined) and control groups. The vertical dashed line indicates the start of treatment.

3.2 Main Treatment Effects

Table 3 reports TWFE estimates from Equation 1 for daily outcomes. Panel A presents results for the first month post-treatment; Panel B extends the estimation window through the second month.

The T4 estimates in Panel A are large and highly significant across all off-peak outcomes. The 40% two-month discount increases the daily probability of any off-peak charging by 0.03 (column 1), which represents a 150% increase relative to the pre-treatment control mean of 0.02. Off-peak minutes rise by 7.14 per day (column 2), a 196% increase over the control mean of 3.65 minutes. Off-peak energy consumption increases by 1,238 kWh (column 3) and off-peak expenditure by 351 KRW (column 4), both significant at the 1% level.

The T2 arm (40%, one month) also generates significant effects, though smaller in magnitude. T2 raises the probability of any off-peak charging by 0.02 (column 1, $p < 0.05$), off-peak energy by 659 kWh (column 3, $p < 0.05$), and off-peak expenditure by 193 KRW (column 4, $p < 0.05$). The T1 and T3 arms (both offering 20% discounts) produce small and statistically insignificant effects, suggesting that a 20% discount is insufficient to induce meaningful behavioral change.

Columns 5–8 report effects on non-off-peak charging. The estimates are generally small and insignificant, with the exception of a marginally significant increase in any non-off-peak charging for T4 (0.02, $p < 0.10$). Importantly, there is no evidence of a reduction in non-off-peak charging minutes or energy, indicating that treatment effects on off-peak charging are additive rather than substitutive. Columns 9–10 confirm this: T4 increases total daily charging minutes by 6.16 ($p < 0.10$) and total kWh by 1,604 ($p < 0.01$).

Panel B extends the estimation window to include two months post-treatment. The T4 effects strengthen: the probability of any off-peak charging remains 0.03 ($p < 0.01$), off-peak minutes rise to 9.58 per day ($p < 0.01$), and off-peak kWh reaches 1,657 ($p < 0.01$). Notably, the T3 arm (20%, two months) now produces marginally significant effects on off-peak minutes (5.37, $p < 0.10$) and significant effects on off-peak kWh (686, $p < 0.05$) and KRW (194, $p < 0.05$), suggesting that a longer exposure period can partially compensate for a smaller discount. The T2 effects attenuate somewhat relative to Panel A, consistent with the incentive having ended after the first month.

The comparison across arms yields two key insights. First, the discount magnitude matters considerably: 40% arms consistently outperform 20% arms, pointing to a nonlinear incentive–response relationship. Second, duration matters: T4 generates larger and more persistent effects than T2, and T3 (two months at 20%) eventually produces effects that T1 (one month at 20%) does not. This is consistent with a habit-formation channel, whereby a longer exposure period allows new charging routines to become established.

Table 3: Treatment Effects on Daily Charging Outcomes

	Off-peak hours				Non-off-peak hours				Total	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	=1 if any	min.	kWh	KRW	=1 if any	min.	kWh	KRW	min.	kWh
Panel A. Up to First Month Post-Treatment, Daily										
T1 × Post	0.01	1.70	383.66	108.10	-0.01	-0.67	-106.89	-76.40	1.03	276.76
	(0.01)	(1.61)	(260.87)	(72.26)	(0.01)	(1.93)	(303.67)	(102.07)	(2.77)	(428.84)
T2 × Post	0.02**	3.09	658.97**	192.83**	0.00	-0.41	101.40	8.30	2.68	760.37
	(0.01)	(1.88)	(293.06)	(80.46)	(0.01)	(1.65)	(291.52)	(101.22)	(2.86)	(481.00)
T3 × Post	0.01	1.67	358.67	100.51	0.02	0.00	376.41	98.66	1.67	735.08*
	(0.01)	(1.71)	(283.16)	(66.51)	(0.01)	(1.98)	(367.73)	(125.81)	(2.70)	(430.06)
T4 × Post	0.03***	7.14***	1238.41***	351.09***	0.02*	-0.98	365.58	189.98	6.16*	1603.99***
	(0.01)	(2.57)	(312.71)	(85.53)	(0.01)	(1.89)	(366.08)	(119.37)	(3.61)	(509.74)
Pre-treat Control: mean	0.02	3.65	535.97	159.26	0.06	6.14	1,533.77	503.09	9.78	2,069.74
Pre-treat Control: SD	0.15	38.32	4,546.15	1,322.29	0.24	39.03	14,395.47	2,396.27	63.72	15,421.74
Observations	230,577	230,577	230,577	230,577	230,577	230,577	230,577	230,577	230,577	230,577
Panel B. Up to Two Months Post-Treatment, Daily										
T1 × Post	0.01	1.99	346.47	93.48	0.01	0.72	191.81	32.33	2.71	538.28
	(0.01)	(2.03)	(255.43)	(78.89)	(0.01)	(1.83)	(286.59)	(97.45)	(3.34)	(439.87)
T2 × Post	0.01*	1.40	474.20	117.32	0.01	1.19	355.47	92.53	2.59	829.68*
	(0.01)	(1.83)	(302.84)	(78.28)	(0.01)	(1.77)	(283.06)	(99.06)	(3.11)	(494.96)
T3 × Post	0.01*	5.37*	685.85**	194.00**	0.02	0.84	420.78	120.58	6.21	1106.63**
	(0.01)	(3.18)	(343.16)	(95.35)	(0.01)	(2.20)	(352.89)	(121.66)	(4.35)	(483.50)
T4 × Post	0.03***	9.58***	1657.04***	454.61***	0.02	-0.63	334.04	177.37	8.94*	1991.08***
	(0.01)	(3.56)	(399.20)	(118.87)	(0.01)	(2.06)	(355.57)	(113.54)	(4.77)	(567.56)
Pre-treat Control: mean	0.02	3.65	535.97	159.26	0.06	6.14	1,533.77	503.09	9.78	2,069.74
Pre-treat Control: SD	0.15	38.32	4,546.15	1,322.29	0.24	39.03	14,395.47	2,396.27	63.72	15,421.74
Observations	243,261	243,261	243,261	243,261	243,261	243,261	243,261	243,261	243,261	243,261

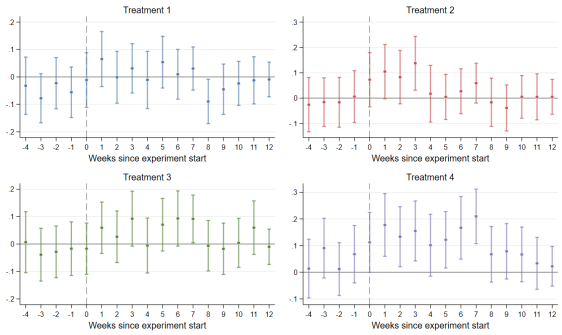
Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regressions include individual and day fixed effects. Standard errors clustered at the individual level in parentheses.

3.3 Event-Study Evidence

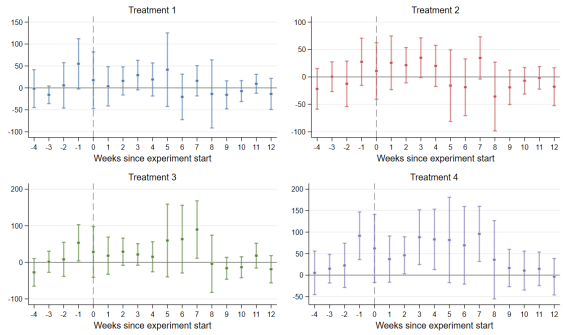
Figure 5 presents event-study estimates at the weekly level for the four primary off-peak outcomes. In each panel, the coefficients for each treatment arm are plotted relative to the week immediately before treatment onset. Three patterns emerge. First, pre-treatment coefficients are close to zero and statistically insignificant for all arms, supporting the parallel-trends assumption. Second, treatment effects for the T4 arm appear promptly in the first post-treatment week and persist throughout the observation window. Third, the T2 effects are visible but smaller, and the 20% arms (T1, T3) show no discernible departure from zero

at any point.

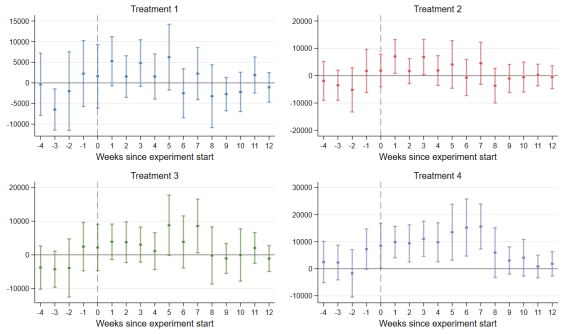
Figure 5: Event-Study Estimates of Off-Peak Charging Outcomes, Weekly



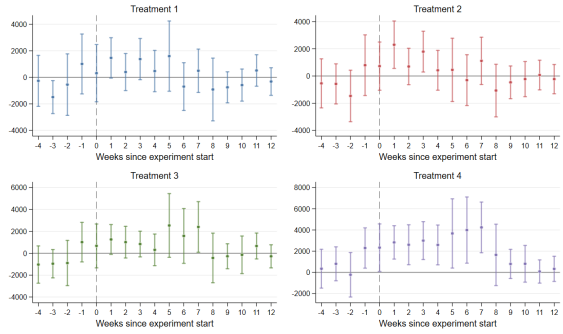
(a) Any off-peak charging



(b) Off-peak charging minutes



(c) Off-peak kWh



(d) Off-peak KRW

Note: Each panel plots weekly event-study coefficients from Equation 1, with the week immediately before treatment onset as the reference period. Shaded bands represent 95% confidence intervals based on standard errors clustered at the individual level.

3.4 Effects by Charger Type

A distinctive feature of our setting is the coexistence of fast and slow chargers, which serve different use cases and operate at different locations. Table 4 decomposes treatment effects by charger type.

In Panel A (first month), the T4 arm generates significant effects on both fast and slow chargers, but the magnitudes differ substantially. On fast chargers, T4 increases the daily probability of off-peak charging by 0.02 ($p < 0.01$) and off-peak minutes by 0.64 ($p < 0.05$). On slow chargers, T4 increases the probability by 0.01 ($p < 0.10$) and off-peak minutes by 6.31 ($p < 0.05$)—roughly ten times the fast-charger effect. Off-peak kWh on slow chargers

rises by 658 ($p < 0.01$) and KRW by 206 ($p < 0.01$). The T2 arm shows a similar pattern: significant effects on fast chargers (probability +0.01, $p < 0.05$; minutes +0.52, $p < 0.10$) but insignificant effects on slow chargers.

Panel B (two months) confirms and amplifies these patterns. T4 effects on slow chargers grow to 8.37 minutes per day ($p < 0.05$) and 797 kWh ($p < 0.05$), while fast-charger effects also strengthen (0.99 minutes, $p < 0.01$; 819 kWh, $p < 0.01$).

The concentration of effects on slow chargers is consistent with overnight home charging as the primary behavioral margin. Shifting slow-charger sessions into the off-peak window requires only a modest adjustment in plug-in timing—for example, waiting until 22:00 rather than plugging in upon arriving home in the early evening. This is a relatively low-cost behavioral change, and the incentive structure of the experiment (discounts conditional on the 22:00–08:00 window) is well-suited to facilitate it.

Table 4: Treatment Effects on Off-Peak Charging Outcomes by Charger Type

	Fast chargers				Slow chargers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	=1 if any	min.	kWh	KRW	=1 if any	min.	kWh	KRW
Panel A. Up to First Month Post-Treatment, Daily								
T1 × Post	0.01	0.23	232.71	62.93	-0.00	1.34	131.23	38.64
	(0.01)	(0.26)	(216.45)	(53.13)	(0.01)	(1.57)	(148.04)	(49.95)
T2 × Post	0.01**	0.52*	376.67*	107.55**	0.01	2.66	288.62	87.62
	(0.01)	(0.29)	(228.00)	(48.88)	(0.01)	(1.84)	(190.00)	(63.85)
T3 × Post	0.01*	0.28	211.78	59.64	0.00	1.40	146.25	41.02
	(0.01)	(0.25)	(222.00)	(41.77)	(0.01)	(1.65)	(161.90)	(50.83)
T4 × Post	0.02***	0.64**	532.01**	126.07***	0.01*	6.31**	657.94***	205.99***
	(0.01)	(0.27)	(227.08)	(46.76)	(0.01)	(2.47)	(231.59)	(73.81)
Pre-treat Control: mean	0.01	0.38	275.81	75.34	0.01	3.10	244.15	78.79
Pre-treat Control: SD	0.10	4.33	3,367.77	873.79	0.10	37.25	2,957.09	960.32
Observations	230,577	230,577	230,577	230,577	230,577	230,577	230,577	230,577
Panel B. Up to Two Months Post-Treatment, Daily								
T1 × Post	0.01	0.22	214.97	54.85	-0.00	1.69	120.88	35.04
	(0.01)	(0.22)	(166.25)	(40.46)	(0.00)	(2.04)	(198.95)	(70.27)
T2 × Post	0.01**	0.56*	421.39	109.44**	0.00	0.75	44.28	6.13
	(0.01)	(0.33)	(260.75)	(51.45)	(0.01)	(1.82)	(177.20)	(63.72)
T3 × Post	0.01**	0.45	358.72	87.93**	0.00	4.87	319.60	103.95
	(0.00)	(0.28)	(231.95)	(43.04)	(0.01)	(3.19)	(262.28)	(88.37)
T4 × Post	0.02***	0.99***	818.63***	181.51***	0.01	8.37**	796.86**	256.86**
	(0.01)	(0.32)	(270.50)	(51.97)	(0.01)	(3.51)	(316.82)	(110.88)
Pre-treat Control: mean	0.01	0.38	275.81	75.34	0.01	3.10	244.15	78.79
Pre-treat Control: SD	0.10	4.33	3,367.77	873.79	0.10	37.25	2,957.09	960.32
Observations	243,261	243,261	243,261	243,261	243,261	243,261	243,261	243,261

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regressions include individual and day fixed effects. Standard errors clustered at the individual level in parentheses.

3.5 Heterogeneity by Charger Access

We next examine whether treatment effects vary with participants' access to charging infrastructure. Table 5 splits the sample by whether the participant has access to a fast charger at home. Among participants with home fast-charger access, T4 generates large and significant

increases in off-peak charging: the probability rises by 0.04 ($p < 0.01$), minutes by 8.20 ($p < 0.05$), and kWh by 1,484 ($p < 0.01$). Interestingly, this group also shows significant increases in non-off-peak charging (any: +0.05, $p < 0.01$; kWh: +1,151, $p < 0.05$), suggesting that the incentive may have raised overall charging engagement rather than merely shifting timing.

Among participants without home fast-charger access, the T4 effects on off-peak outcomes remain significant (kWh: +1,029, $p < 0.05$; KRW: +246, $p < 0.05$) but are accompanied by a significant reduction in non-off-peak charging minutes (-6.15, $p < 0.05$). This pattern is consistent with substitution: participants who rely on shared or public chargers appear to shift existing charging sessions to off-peak hours rather than adding new ones. The T2 arm (40%, one month) also produces significant off-peak effects in this subgroup (any: +0.03, $p < 0.05$; kWh: +932, $p < 0.05$), while T1 shows a significant decrease in non-off-peak charging.

Table 5: Treatment Effects by Home Charger Access: Up to First Month Post-Treatment

	Off-peak hours				Non-off-peak hours				Total	
	(1) =1 if any	(2) min.	(3) kWh	(4) KRW	(5) =1 if any	(6) min.	(7) kWh	(8) KRW	(9) min.	(10) kWh
Home Charger Access: = 1 if any fast charger										
T1 × Post	0.01	1.35	236.14	105.73	0.03**	4.49	817.49**	215.32*	5.84*	1053.63**
	(0.01)	(1.42)	(328.22)	(94.43)	(0.01)	(2.73)	(376.20)	(118.90)	(3.43)	(533.67)
T2 × Post	0.01	2.76	421.76	150.52	0.01	-0.51	300.05	56.00	2.25	721.81
	(0.01)	(2.84)	(378.15)	(105.16)	(0.02)	(2.10)	(396.85)	(133.71)	(3.87)	(617.99)
T3 × Post	0.01	0.26	466.51	119.15	0.03*	1.87	742.84	212.63	2.13	1209.35*
	(0.01)	(1.95)	(516.54)	(107.60)	(0.02)	(1.78)	(608.07)	(206.66)	(2.66)	(662.78)
T4 × Post	0.04***	8.20**	1483.59***	459.42***	0.05***	3.50	1151.13**	475.08**	11.70**	2634.72***
	(0.01)	(3.86)	(460.36)	(121.81)	(0.02)	(2.57)	(582.06)	(184.26)	(5.69)	(789.39)
Pre-Treat Control: mean	0.02	3.41	526.64	147.40	0.07	6.21	1,553.37	489.26	9.62	2,080.01
Pre-Treat Control: SD	0.15	38.18	4,495.69	1,194.55	0.25	39.56	18,069.84	2,349.92	63.46	18,820.56
Observations	117,070	117,070	117,070	117,070	117,070	117,070	117,070	117,070	117,070	117,070
Home Charger Access: no fast charger										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
T1 × Post	0.00	2.02	564.49	109.95	-0.05***	-7.00***	-1262.95***	-441.41***	-4.98	-698.46
	(0.01)	(3.06)	(408.86)	(111.18)	(0.02)	(2.62)	(461.51)	(165.16)	(4.52)	(671.10)
T2 × Post	0.03**	3.37	932.30**	237.35*	-0.01	-1.19	-286.61	-99.93	2.18	645.69
	(0.02)	(2.58)	(443.97)	(123.32)	(0.02)	(2.58)	(411.53)	(150.03)	(4.43)	(726.77)
T3 × Post	0.01	2.66	301.23	82.63	-0.00	-2.74	-197.21	-81.43	-0.07	104.02
	(0.01)	(2.77)	(274.38)	(83.84)	(0.02)	(3.42)	(439.51)	(155.74)	(4.72)	(558.19)
T4 × Post	0.02	5.99*	1028.65**	245.75**	-0.01	-6.15**	-587.33	-146.66	-0.17	441.32
	(0.01)	(3.47)	(411.99)	(118.81)	(0.02)	(2.80)	(437.51)	(151.50)	(4.69)	(633.23)
Pre-Treat Control: mean	0.02	3.96	548.32	174.97	0.06	6.04	1,507.81	521.40	9.99	2,056.13
Pre-Treat Control: SD	0.15	38.50	4,612.24	1,474.47	0.24	38.31	7,012.72	2,456.28	64.05	9,145.06
Observations	112,998	112,998	112,998	112,998	112,998	112,998	112,998	112,998	112,998	112,998

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regressions include individual and day fixed effects. Standard errors clustered at the individual level in parentheses.

Table 6 conducts the analogous exercise for workplace charger access. Additional heterogeneity results, including splits by age group, pre-treatment charger-type usage, and day of week, are reported in Appendix A.

Table 6: Treatment Effects by Workplace Charger Access: Up to First Month Post-Treatment

	Off-peak hours				Non-off-peak hours				Total	
	(1) =1 if any	(2) min.	(3) kWh	(4) KRW	(5) =1 if any	(6) min.	(7) kWh	(8) KRW	(9) min.	(10) kWh
Workplace Charger Access: = 1 if any fast charger										
T1 × Post	0.02	0.75	446.98	116.98	0.01	-0.73	-43.41	-59.98	0.02	403.57
	(0.01)	(1.34)	(382.71)	(106.93)	(0.02)	(3.56)	(535.33)	(183.59)	(4.45)	(718.11)
T2 × Post	-0.00	0.54	-38.02	14.11	-0.01	-3.00	-138.92	-52.42	-2.46	-176.94
	(0.01)	(1.62)	(260.16)	(80.99)	(0.02)	(2.88)	(522.20)	(185.80)	(3.90)	(667.83)
T3 × Post	0.01	2.16	178.09	60.01	0.01	-4.77	46.54	5.35	-2.61	224.63
	(0.02)	(3.67)	(420.59)	(120.30)	(0.02)	(3.58)	(546.82)	(192.31)	(5.42)	(752.22)
T4 × Post	0.05***	11.05**	2001.37***	548.53***	0.03	-1.50	556.88	191.90	9.55	2558.25***
	(0.01)	(4.35)	(510.97)	(136.01)	(0.02)	(3.15)	(561.34)	(188.14)	(6.65)	(832.78)
Pre-Treat Control: mean	0.03	3.34	633.87	198.89	0.07	7.04	1,761.52	603.43	10.38	2,395.39
Pre-Treat Control: SD	0.16	32.09	4,816.82	1,507.19	0.26	40.86	7,782.01	2,687.06	60.91	9,853.29
Observations	93,147	93,147	93,147	93,147	93,147	93,147	93,147	93,147	93,147	93,147
Workplace Charger Access: no fast charger										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
T1 × Post	0.00	2.41	334.78	101.70	-0.02	-0.52	-160.40	-91.62	1.89	174.38
	(0.01)	(2.64)	(354.65)	(97.31)	(0.02)	(1.99)	(339.03)	(109.37)	(3.47)	(511.85)
T2 × Post	0.04***	4.88	1146.64**	319.25***	0.01	1.55	249.16	41.88	6.42	1395.80**
	(0.01)	(2.97)	(450.96)	(121.20)	(0.01)	(1.92)	(322.54)	(106.91)	(3.95)	(642.50)
T3 × Post	0.01	1.41	437.38	121.30	0.02	2.76	494.02	123.96	4.17	931.40*
	(0.01)	(1.89)	(364.45)	(77.66)	(0.02)	(2.30)	(454.90)	(151.41)	(2.92)	(488.16)
T4 × Post	0.02	4.27	680.16*	207.43**	0.02	-0.45	217.96	184.38	3.82	898.12
	(0.01)	(3.06)	(372.79)	(104.27)	(0.02)	(2.27)	(475.46)	(150.47)	(3.80)	(613.40)
Pre-Treat Control: mean	0.02	3.90	455.34	126.62	0.06	5.39	1,346.21	420.45	9.29	1,801.56
Pre-Treat Control: SD	0.14	42.77	4,308.92	1,146.87	0.23	37.44	18,109.27	2,123.67	65.94	18,803.87
Observations	136,921	136,921	136,921	136,921	136,921	136,921	136,921	136,921	136,921	136,921

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regressions include individual and day fixed effects. Standard errors clustered at the individual level in parentheses.

Table 7: Treatment Effects by Pre-Treatment Charger-Type Usage Frequency: Up to First Month Post-Treatment

	Fast chargers				Slow chargers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	=1 if any	min. rate	kWh rate	KRW rate	=1 if any	min. rate	kWh rate	KRW rate
Panel A. High Fast Session Rate Users								
T1 × Post	0.024	0.024	0.024	0.024	-0.003	-0.003	-0.003	-0.003
	(0.018)	(0.018)	(0.018)	(0.018)	(0.006)	(0.006)	(0.006)	(0.006)
T2 × Post	0.035**	0.035**	0.035**	0.035**	0.001	0.001	0.001	0.001
	(0.015)	(0.015)	(0.015)	(0.015)	(0.008)	(0.008)	(0.008)	(0.008)
T3 × Post	0.022*	0.022*	0.022*	0.022*	-0.004	-0.004	-0.005	-0.005
	(0.013)	(0.013)	(0.013)	(0.013)	(0.007)	(0.007)	(0.006)	(0.006)
T4 × Post	0.040***	0.040***	0.040***	0.040***	-0.002	-0.002	-0.002	-0.002
	(0.015)	(0.015)	(0.015)	(0.015)	(0.007)	(0.007)	(0.007)	(0.007)
Pre-treat Control: mean	0.022	0.022	0.022	0.022	0.003	0.003	0.003	0.003
Pre-treat Control: SD	0.148	0.147	0.147	0.147	0.057	0.052	0.051	0.051
Observations	89,584	89,584	89,584	89,584	89,584	89,584	89,584	89,584
Panel B. High Slow Session Rate Users								
T1 × Post	0.002	0.001	0.001	0.001	-0.001	-0.001	-0.001	-0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.014)	(0.014)	(0.014)	(0.014)
T2 × Post	0.002***	0.002***	0.002***	0.002***	0.008	0.008	0.007	0.007
	(0.001)	(0.001)	(0.001)	(0.001)	(0.019)	(0.019)	(0.019)	(0.019)
T3 × Post	0.004**	0.003*	0.004*	0.004*	0.010	0.010	0.010	0.010
	(0.002)	(0.002)	(0.002)	(0.002)	(0.016)	(0.016)	(0.016)	(0.016)
T4 × Post	0.007**	0.006*	0.006*	0.006*	0.024	0.024	0.024	0.024
	(0.003)	(0.003)	(0.003)	(0.003)	(0.016)	(0.016)	(0.016)	(0.016)
Pre-treat Control: mean	0.002	0.002	0.002	0.002	0.036	0.036	0.036	0.036
Pre-treat Control: SD	0.044	0.042	0.042	0.042	0.185	0.185	0.185	0.185
Observations	89,584	89,584	89,584	89,584	89,584	89,584	89,584	89,584

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All regressions include individual and day fixed effects. Standard errors clustered at the individual level in parentheses.

4 Discussion

4.1 Comparing Korean Residential and Californian Workplace Charging Incentives

One recent field experiments offer a natural counterpoint to our findings. For example, Garg et al. (2025) run a randomized intervention in a large workplace charging network at UCSD, in which 629 drivers receive flat (that is, time-invariant) price discounts of approximately 50% or 75% off the base rate, alongside an environmental nudge highlighting the climate benefits of midday charging. The empirical setting is solar-abundant: California's grid produces lowest-marginal-emission electricity during the late-morning and midday hours when employees are at work, so a successful policy should pull charging into the solar peak.

Conversely, our setting is residential and fossil-backed at night: most Korean charging occurs through apartment-based slow chargers, the dominant overnight fuel mix is nuclear and coal rather than solar, and the price signal is conditional on the off-peak window of 22:00–08:00. The two studies therefore vary instrument design (flat versus time-conditional), behavioral margin (workplace daytime versus residential overnight), lived environment characteristics (comparatively, sparse and less dense versus concentrated and highly dense populations), and grid composition (solar-abundant versus fossil-overnight) simultaneously, which makes the comparison instructive and complimentary.

Garg et al. (2025) report that flat workplace discounts shift charging away from the solar peak. The first phase of their financial intervention raises early-morning sessions by 4.9% and overnight sessions by 3.3%; the second phase shifts evening sessions upward by 12.9%. They attribute this counterintuitive pattern to perceived charger scarcity: when drivers expect that a flat discount will pull additional vehicles into a congested workplace network, they re-time their arrivals away from peak-utilization hours to secure access. A follow-up scarcity experiment confirms the mechanism. The environmental implication is unfavorable: the second financial discount in Garg et al. (2025) increases CO_2 emissions by approximately 1.94% per driver because charging is pulled into higher-marginal-emission hours. Their environmental nudge, in contrast, modestly re-times charging into the solar peak. In a

solar-abundant grid, information dominates price as a temporal-shifting instrument.

Our findings stand in instructive contrast: A 40% discount conditional on the off-peak window raises the daily probability of off-peak charging by 150% and off-peak energy by 196% relative to the control mean, with effects concentrated on residential slow chargers—the dominant Korean charging margin. A 20% discount of the same form produces small, primarily insignificant short-run effects, suggesting a nonlinear price-response relationship. Translated into an arc elasticity, our pooled 40%-versus-20% comparison yields $\sigma \approx -1.45$ for off-peak energy, close to Bailey et al. (2025b) and Bernard et al. (2025) for residential and public charging, and roughly three times the -0.46 reported by Garg et al. (2025) for workplace flat discounts.

Two differences appear to drive the gap. First, our discount is conditional on the off-peak window, so the price signal points drivers toward a specific time of day rather than toward the workplace as a location. Second, residential overnight slow charging is a low-friction behavioral margin: shifting a plug-in time by one or two hours requires no schedule disruption, in contrast to moving a parked workplace vehicle during the workday.

Overall, the two experiments suggest that the environmental sign of an EV-charging incentive is jointly determined by instrument design and the dominant charging margin, rather than by either feature alone. A flat discount applied to a congested daytime network pulls drivers out of the solar peak in a solar-abundant grid; a time-conditional discount applied to a residential overnight margin pulls drivers into the off-peak window in a fossil-backed overnight grid. The two cases are not contradictory but complementary: each isolates a corner of a broader policy-design space in which time-conditioning, congestion, and grid composition interact.

Two implications follow. First, policymakers should match the design of the price signal to the dominant charging margin in their jurisdiction; flat discounts that work for residential bill-shifting can backfire in shared workplace networks, and time-conditional discounts that work in residential overnight settings may be impractical where chargers are bundled with parking. Second, future experimental work should hold instrument design fixed while varying grid composition, or vice versa, so that the comparative elasticity surface that Bailey et al. (2025b), Bernard et al. (2025), Garg et al. (2025), and the present paper are beginning to

map out can be filled in along its remaining dimensions.

4.2 Elasticity Calculation

Although a growing experimental literature has begun to estimate the price elasticity of EV-charging demand across diverse settings, no comparable estimate yet exists for an Asian economy with predominantly apartment-based slow charging. We address this gap by translating our intent-to-treat estimates into an arc price elasticity of off-peak charging. Pooling the 40%-off and 20%-off arms over the two-month window and applying the midpoint convention, we obtain $\sigma \approx -1.45$ for off-peak energy (See the Appendix more in detail), which places the Korean residential setting close to the -1.59 reported for residential off-peak charging in Calgary (Bailey et al. 2025b) and the -1.45 to -1.00 reported for public dynamic pricing in the United Kingdom (Bernard et al. 2025), and well above the -0.46 reported for workplace flat discounts at UCSD (Garg et al. 2025). The within-duration comparisons further suggest that this elasticity is not realized instantaneously: σ rises from approximately -0.91 in the one-month arms to approximately -1.99 in the two-month arms, consistent with a habit-formation channel in which drivers internalize the discount only as the off-peak charging routine becomes habitual.

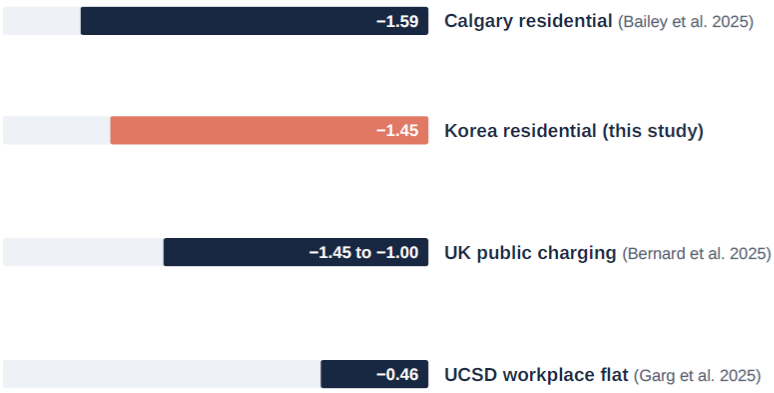


Figure 6: Off-peak EV charging price elasticity (arc, midpoint)

Note: Arc price elasticity of off-peak EV-charging demand across recent field experiments. The Korean residential setting sits close to Calgary residential and UK public-charging estimates, and well above the workplace-flat-discount elasticity from UCSD. Within our study, rises from approximately 0.91 in the one-month arms to 1.99 in the two-month arms, pointing to a habit-formation channel.

Three implications follow. First, residential time-conditional discounts appear to reach a more elastic behavioral margin—overnight slow charging at home—than workplace flat discounts can mobilize, which is consistent with the broader demand-response literature in which the peak-to-off-peak price ratio is the dominant predictor of measured response (Faruqui and Sergici 2017). Second, the duration of exposure matters as much as the magnitude of the discount, indicating that a one-month rate experiment is likely to understate the long-run elasticity of off-peak charging. Third, the comparison across settings underscores that policymakers should match the design of the price signal to the dominant charging margin in their jurisdiction; in dense, apartment-based markets such as Korea, time-conditional discounts on residential overnight charging tend to move behavior in ways that flat workplace discounts in solar-abundant grids cannot.

4.3 Environmental Impact and Fiscal Abatement Cost

A central question for any charging-incentive program is whether the temporal shift it induces actually reduces grid-related emissions, and at what fiscal cost per ton of CO_2 avoided. Although our intervention raises off-peak charging substantially, the environmental sign of this shift depends on the marginal carbon intensity of the Korean grid in the incentivized hours relative to the displaced hours. We adapt the empirical framework of Garg et al. (2025) to the Korean residential setting, using hourly generation-by-fuel data from the Korea Power Exchange (KPX EPSIS) for 2024 [1] to construct three operationally distinct carbon-intensity (CI) series, and compute the implied change in CO_2 emissions and the fiscal abatement cost per ton of CO_2 for each treatment arm.

Framework. Following Garg et al. (2025), we decompose the per-user change in CO_2 from each treatment arm $k \in \{T2, T3, T4\}$ into a within-block timing component:

$$\Delta \mathrm{CO}_{2,k} \;=\; \beta_k^{\mathrm{off}} \cdot \mathrm{CI}^{\mathrm{off}} \;+\; \beta_k^{\mathrm{non}} \cdot \mathrm{CI}^{\mathrm{non}} \tag{2}$$

[1] KPX EPSIS 2025 data will be released on September 1, 2026.

where β_k^{off} and β_k^{non} are the per-user-day treatment-effect coefficients on energy charged in off-peak and non-off-peak windows, and CI^{off} and CI^{non} are the corresponding marginal CIs. The fiscal abatement cost per ton of CO_2 is the per-user discount expenditure divided by the avoided emissions, monetized at the social cost of carbon (SCC).

Korean marginal carbon intensity from KPX 2024 dispatch data. We use the KPX EPSIS hourly dataset of generation by fuel source for 2024 (8,784 hours × 26 fuel categories) to construct three CI methodologies that bracket the marginal-CI question. (1) The average grid CI weights all generation, including zero-emission nuclear and renewables, by hour-of-day; this is the consumption-weighted CI relevant for accounting purposes but not for marginal-impact analysis. (2) The operating margin CI (UNFCCC CDM Executive Board 2023) excludes must-run baseload and renewables, computing the emission-weighted average across only fossil-dispatched generation; this is the standard proxy for the marginal generator's CI. (3) The marginal-fuel CI assigns the emission factor of the highest-EF actively-dispatched fossil source in each hour, providing an upper bound on what the marginal MW of demand would emit. Emission factors follow IPCC AR6 lifecycle ranges and IEA Emissions Factors 2024 conventions: 820 gCO_2/kWh for Korean bituminous coal, 950 for anthracite, 750 for IGCC, 400 for LNG combined-cycle, and zero for nuclear, hydro, wind, solar, and biogenic sources.

Figure 7 (panel a) plots the three-hourly CI series. Two patterns are notable. First, all three measures are remarkably flat across the 24-hour cycle: the operating margin CI moves between roughly 584 and 640 gCO_2/kWh, and the marginal-fuel CI is virtually constant at approximately 818 gCO_2/kWh because Korean coal generation runs nearly around the clock without ever reaching full system-wide capacity. Second, although the variation is small, the off-peak block (22:00–08:00) does carry a higher operating margin CI than the non-off-peak block (08:00–22:00): off-peak averages 624 gCO_2/kWh against non-off-peak 592, a spread of approximately +32 gCO_2/kWh. This pattern reverses the conventional intuition based on average grid CI (under which overnight hours look cleaner because nuclear dominates baseload and demand is low) and is consistent with the Powell et al. (2022) finding that off-peak charging in coal-backed systems can compound rather than alleviate generation-

level emissions, although the magnitude of the effect in Korea is small relative to what their stylized model implied.

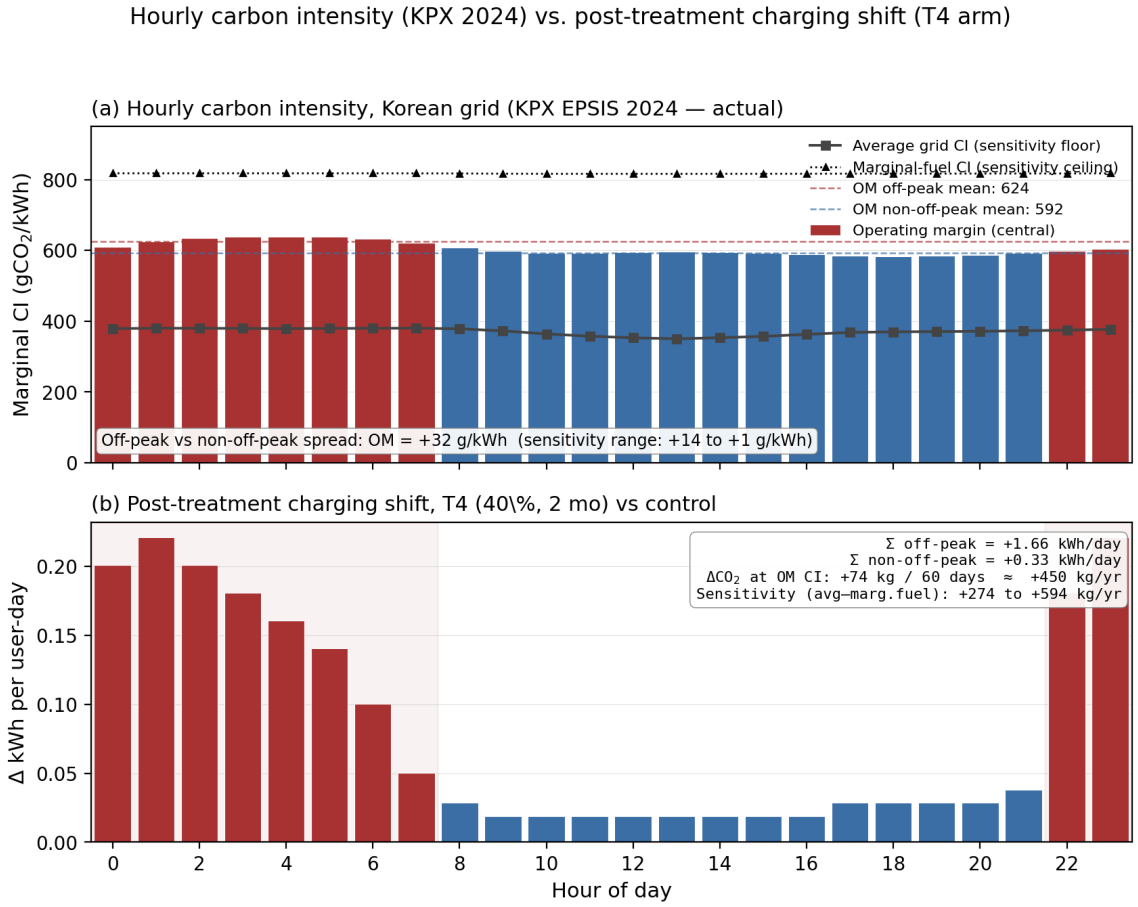


Figure 7: Hourly carbon intensity from KPX EPSIS 2024 dispatch data (panel a) and post-treatment charging shift for the T4 arm relative to control (panel b). Panel (a) plots three CI methodologies: operating margin (red/blue bars, central estimate), average grid CI (line with squares, sensitivity floor), and marginal-fuel CI (dotted line, sensitivity ceiling). Off-peak hours (22:00–08:00) are shaded in red; non-off-peak in blue. The operating-margin spread between off-peak and non-off-peak blocks is approximately +32 gCO_2/kWh.

CO_2 effect under three counterfactuals. Because our data record only platform-observed charging, the empirical estimand admits three distinct counterfactuals, each yielding a different environmental impact. *Scenario A* (pure additive): the off-peak increase reflects net new charging on the Korean grid; the relevant counterfactual is no charging at all, and equation (2) applies directly. *Scenario B* (within-platform substitution): the off-peak increase is fully offset by a reduction in non-off-peak charging by the same user on

the same platform, so $\Delta CO_{2,k} = \beta_k^{off} \cdot (CI^{off} - CI^{non})$. *Scenario C* (cross-platform substitution): the off-peak increase reflects users moving sessions from a different operator that also charged off-peak, so there is no temporal shift in real terms and $\Delta CO_{2,k} = 0$. The empirical results—no consistent reduction in non-off-peak charging on the platform alongside the inability to observe other operators—are consistent with a mixture of Scenarios A and C, and the true environmental impact is bounded between the two.

Headline results. Table 8 reports the implied CO_2 change and fiscal abatement cost for T2, T3, and T4 over the active treatment window, evaluated at the operating margin CI and SCC = \$185/tCO_2 (Garg et al. 2025). Three findings stand out. First, under both Scenario A and Scenario B, all three significant treatment arms increase CO_2 emissions, because the operating margin CI of the incentivized off-peak hours exceeds that of the displaced hours. For T4, the additive-scenario impact is approximately +74 kg CO_2 per user over the 60-day treatment window, or approximately +450 kg per user-year if the behavioral effect persists; the within-platform substitution scenario implies a smaller +3 kg per user over the 60-day window because the off-peak/non-off-peak spread in the operating margin CI is only +32 gCO_2/kWh. Second, the fiscal abatement cost is negative across arms and scenarios under the additive counterfactual: in all cases the program transfers money to drivers in exchange for additional emissions. For T4 under Scenario A, the implied figure is approximately −\$41 per ton CO_2—the program pays \$41 per ton of added emissions, an inverted abatement cost. Third, the magnitudes scale monotonically with both discount intensity and duration.

Table 8: CO_2 effect, fiscal cost, and abatement cost per treatment arm (KPX 2024 operating margin CI, SCC = \$185)

Arm	Days	ΔE^{off} (kWh)	ΔE^{non} (kWh)	ΔCO_2 (A) (kg/user)	ΔCO_2 (B) (kg/user)	Fiscal cost (USD/user)	\$/tCO_2 (A)
T2 (40%, 1 mo)	30	19.8	3.0	+14.2	+0.6	0.83	−\$59
T3 (20%, 2 mo)	60	41.2	25.3	+40.7	+1.3	0.85	−\$21
T4 (40%, 2 mo)	60	99.4	20.0	+74.0	+3.2	3.06	−\$41

Notes: Values follow equation (2) using CI^{off} = 624 gCO_2/kWh and CI^{non} = 592 gCO_2/kWh, derived from KPX EPSIS 2024 hourly dispatch using the operating-margin methodology (UNFCCC CDM Executive Board 2023). Scenario A treats the off-peak shift as an additive new charging on the platform; Scenario B treats it as a within-platform substitution from non-off-peak. Fiscal cost is the per-user discount transferred over the active window. \$/tCO_2 negative indicates that the program pays for additional emissions rather than abating them.

Sensitivity to CI methodology and SCC. Table 9 reports the T4 environmental impact across the three CI methodologies and three SCC values. The annualized emission increase ranges from approximately +274 kg/user/year (using average grid CI) to +594 kg (using marginal-fuel CI), with the operating-margin central estimate at +450 kg. Across all nine cells of the CI × SCC table, CO_2 is positive: the qualitative result that the program adds rather than abates emissions is robust to reasonable variation in the CI assumption. The associated social damage at SCC = \$185 ranges from \$50.66 to \$109.93 per user-year.

Table 9: T4 environmental impact: sensitivity to CI methodology and SCC (additive, annualized)

CI methodology	ΔCO_2 (kg/user-yr)	SCC = \$51	SCC = \$185	SCC = \$210
Average grid CI	+274	\$13.96	\$50.66	\$57.51
Operating margin CI (central)	+450	\$22.95	\$83.23	\$94.49
Marginal-fuel CI	+594	\$30.30	\$109.93	\$124.79

Notes: CI methodologies derived from KPX EPSIS 2024: average CI weights all dispatched generation including nuclear and renewables; operating margin CI weights only fossil-dispatched generation (UNFCCC ACM0002); marginal-fuel CI takes the emission factor of the highest-EF actively-dispatched fossil source. SCC values: EPA (2016) interim, EPA (2023) updated, and the \$210 value used in (Garg et al. 2025). Emission impact computed under Scenario A (pure additive) and annualized assuming the post-treatment behavioral effect persists.

Comparison with Garg et al. (2025). The environmental sign of our intervention echoes the result (Garg et al. 2025) obtained for the prolonged "Discount 2" phase of their UCSD workplace experiment, which also increases CO_2 emissions per driver because the discount pulls charging away from the solar peak in a solar-abundant grid. The mechanisms differ. In the Californian setting, flat workplace discounts shift charging out of the solar window through a perceived-scarcity channel that is specific to congested workplace networks. In the Korean residential setting, time-conditional discounts shift charging into the off-peak window as designed; the perverse environmental sign arises not from instrument failure but from a mismatch between the policy's hour target and the hours at which the marginal generator is dirtier. Magnitudes also differ: Garg's prolonged discount increases CO_2 by approximately 1.94% per driver, while our T4 estimate of +450 kg/user-year corresponds to approximately a doubling of the participant's charging-related emissions in the additive scenario. The two studies together suggest that the environmental sign of an EV-charging incentive is jointly determined by instrument design and grid composition: a flat discount pulls charging away

from clean midday hours in solar-abundant grids, while a time-conditional discount pulls charging into hours of higher operating-margin CI in coal-backed grids.

Caveats. Three caveats apply. First, our marginal CI series uses 2024 dispatch as a proxy for the August–October 2025 experiment window; the Korean fuel mix is evolving rapidly, with solar capacity expanding and coal retirements ongoing, so the relevant marginal CI in 2025–2026 may compress further. Second, the cross-platform substitution scenario—which yields zero environmental impact—cannot be ruled out with platform-only data; a follow-up survey or odometer-based verification along the lines of (Garg et al. 2025) would help distinguish Scenarios A and C. Third, the off-peak/non-off-peak operating-margin spread of +32 gCO_2/kWh is small in absolute terms, and the headline "policy backfires" finding rests on the persistence of this spread; if Korean dispatch flattens further as renewables grow, the environmental case against the off-peak discount weakens further (though it does not become favorable, because solar generation does not displace coal during overnight hours in the foreseeable Korean grid).

4.4 Welfare and fiscal cost of the discount

The CO_2 accounting in Section 4.3 treats the per-user discount expenditure as a uniform cost. From a welfare standpoint, however, this expenditure is heterogeneous: part of it is a transfer from KEPCO and Korean taxpayers to participating drivers, and part is a real economic cost arising from electricity consumed at a price below its social marginal cost (SMC). Garg et al. (2025) make precisely this distinction for the Californian workplace setting and conclude, drawing on (Borenstein and Bushnell 2022), that retail electricity prices in California exceed the social marginal cost by a factor of roughly two, so that workplace discounts on top of the base rate function almost entirely as transfers. We perform the analogous decomposition for Korea and find a sharply different pattern: the off-peak EV charging tariff that anchors our experiment is already approximately at-cost, and a percentage discount on top of it pushes price below SMC, introducing a new welfare distortion rather than correcting an existing one.

The Korean institutional setting helps explain the contrast. KEPCO is a vertically-

integrated state-owned utility whose retail tariffs are set by the Ministry of Trade, Industry, and Energy. The corporation accumulated approximately 206 trillion KRW of debt as of mid-2024 and recorded combined operating losses of roughly 43 trillion KRW between 2021 and 2023, primarily because retail tariffs were held below wholesale costs in the wake of the post-2021 energy-price surge (M. C. Kim 2024). The off-peak EV charging tariff that anchors our experiment is set at 76.85 KRW/kWh (low-voltage, weighted average across August–October 2025), itself well below the residential progressive tariff (105–244 KRW/kWh across the three-tier structure) and roughly comparable to wholesale costs during overnight hours. The Korea Power Exchange wholesale system marginal price (SMP) averaged approximately 126 KRW/kWh across all hours in 2024, with off-peak SMP plausibly closer to 90 KRW/kWh because LNG combined-cycle units are dispatched down overnight; Korean coal-fired plants set the off-peak margin at a short-run marginal cost (SRMC, fuel plus variable operations and maintenance) of approximately 65 KRW/kWh. The pre-discount retail price, therefore, sits between SRMC and SMP. The 40% T4 discount sets the post-discount price at 46.11 KRW/kWh, which is below both. This is structurally different from the Californian setting in (Garg et al. 2025), where the base workplace rate of \$0.30/kWh is roughly four times the wholesale locational marginal price of \$0.07/kWh, and even the 75% discount remains at or above the SMC.

Decomposing the per-user discount expenditure for T4 over the active 60-day treatment window into a fiscal transfer (the discount fraction multiplied by the off-peak retail price and the total off-peak energy), a resource cost (the additional kWh multiplied by SRMC, under the additive counterfactual), and a deadweight loss (the standard linear-demand triangle on the additional kWh, $\frac{1}{2} \cdot \beta_{T4}^{\text{off}} \cdot (\text{SMP} - p^{\text{post}})$), we find a fiscal transfer of approximately \$3.06 per user, a resource cost of approximately \$4.90, and a deadweight loss of approximately \$1.65, for a transfer share of roughly 65%. The transfer share is somewhat smaller than the figure that would obtain in California, where the discount is almost entirely a transfer; the deadweight-loss component scales with discount intensity, ranging from \$0.12 per user under T1 (20%, one month) to \$1.65 under T4 (40%, two months). Sensitivity analysis indicates that the transfer share for T4 ranges from roughly 56% to 74% across plausible SMP values of 75–110 KRW/kWh, so the qualitative picture is robust to reasonable variation in the

wholesale-cost benchmark.

If the T4 discount were extended to all Korean EV drivers with apartment-based slow chargers—approximately 70% of the fleet—and behavioral effects scaled linearly, the annual fiscal transfer would be on the order of 9 to 13 billion KRW, or roughly 7 to 10 million USD per year, rising to approximately 14 million USD per year at full-fleet hypothetical coverage in 2026 (Ministry of Land, Infrastructure and Transport 2026). These magnitudes are modest relative to KEPCO's accumulated debt of roughly $140 billion, but two distributional considerations make them non-trivial. First, KEPCO recovers its losses primarily from industrial ratepayers and from periodic government capital injections, so the discount cross-subsidizes EV-driving households—who are on average wealthier than the population mean—from non-EV-owning industrial consumers and taxpayers. Second, the discount layered on top of an already-low EV charging tariff compounds an existing implicit subsidy, so the marginal welfare cost of additional discount intensity rises as the underlying tariff is squeezed further below cost.

Although fiscal discounts on off-peak EV charging are administratively simple and politically attractive, our results suggest that in jurisdictions where the underlying tariff already sits at or below SMC, percentage discounts add fiscal cost while introducing welfare loss on the marginal kWh consumed. Korean policymakers should therefore consider three complementary moves. First, the welfare cost of the discount should be benchmarked against alternative instruments—managed-charging systems, smart-charger timer subsidies, or solar-aligned midday discounts—whose welfare consequences may be more favorable under Korean tariff conditions. Second, any future scale-up of off-peak discounts should be paired with a reform that aligns the underlying EV charging tariff more closely with hourly SMC, so that the discount is not layered on top of an already-distorted base. Third, the distributional incidence of the program should be made explicit alongside the environmental implications discussed in Section 4.3, particularly given that the welfare cost falls on a different population from the welfare benefit.

4.5 Implications

Our findings provide implications on the design of off-peak charging financial incentives, on the environmental impact of EV charging in Korea, and on the welfare consequences of layering discounts on top of an already-low EV charging tariff.

First, our nonlinear incentive–response relationship is the central empirical finding of this study. A 40% discount crosses a behavioral threshold that a 20% discount does not, consistent with the broader demand-response literature finding that the peak-to-off-peak price ratio must be sufficiently large to overcome inertia. Translating the intent-to-treat estimates into an arc price elasticity, as derived in Section 4.2, yields $\sigma \approx -1.45$ for off-peak energy in the pooled two-month window; the within-duration comparisons further show that this elasticity is not realized instantaneously but rises from approximately -0.91 in the one-month arms to approximately -1.99 in the two-month arms, indicating that a one-month rate experiment is likely to understate the long-run response of off-peak charging.

Second, electricity-related emissions in Korea are relatively lower overnight on an average basis, when nuclear and coal together dominate baseload, but they are not as clean as in nuclear-dominant overnight grids. Total emissions are highest during daytime and evening peaks because LNG, the marginal non-base-load fuel, is dispatched at scale; however, the *marginal* carbon intensity of the next kWh of demand is comparable across off-peak and non-off-peak hours, because Korean coal generation runs near-baseload around the clock and rarely reaches full system-wide capacity. Using KPX EPSIS 2024 hourly dispatch data, we find that the operating-margin carbon intensity averages approximately 624 gCO_2/kWh in the off-peak window (22:00–08:00) against 592 gCO_2/kWh in the non-off-peak window—a spread of +32 gCO_2/kWh that is smaller than common stylized estimates suggest but is consistently positive across alternative CI methodologies. The evening "solar cliff" that sharpens this pattern in high-solar systems is currently a minor factor in Korea, though it will grow as solar capacity rapidly expands. As solar penetration grows in the future, midday charging on solar surplus could become preferable to overnight charging.

The back-of-envelope concern raised in earlier drafts of this section is now formalized in the quantitative decomposition reported in Section 4.3. Because about 57% of the response

is additive rather than substitutive on the platform we observe, and because the relevant counterfactual cannot be empirically distinguished from cross-platform substitution, the net effect on system-level emissions is bounded between approximately zero (cross-platform substitution) and approximately +450 kg CO_2 per user-year (pure additive at operating-margin CI). Powell et al. (2022)'s finding reinforces this result: off-peak incentives that push residential load further into the overnight coal-backed window may compound rather than alleviate generation-level grid stress, and expanding daytime charging infrastructure represents a more structurally sound long-run complement to price-based demand shifting.

Third, the concentration of treatment effects on slow chargers, and the divergence between participants with and without home fast-charger access, suggest the importance of home slow chargers for residential electricity consumption patterns. Because about 90% of Korea's EV chargers are slow chargers and these are predominantly located in residential settings, the behavioral margin that the experiment identifies as most elastic is also the most scalable.

Fourth, although time-of-use incentives are often presented as climate-aligned policy tools, our results suggest that in coal-backed overnight grids, off-peak charging discounts may shift load into hours where the marginal generator is dirtier, generating a negative environmental return on the public expenditure—approximately −\$41 per ton of CO_2 *added* under SCC = \$185/tCO_2, an inverted abatement cost. The welfare picture is similarly unfavorable along a second dimension, as Section 4.4 documents. Because the Korean off-peak EV charging tariff is already approximately at-cost—narrowly above the short-run marginal cost of overnight coal generation and narrowly below the wholesale system marginal price—a 40% discount layered on top of this rate pushes the marginal kWh price below social marginal cost and introduces a small but non-trivial deadweight loss on the additional consumption, in addition to the fiscal transfer from KEPCO and Korean taxpayers to participating drivers.

Korean policymakers should therefore complement off-peak price signals with at least three structural changes. First, the off-peak window should be aligned with the diurnal cycle of marginal carbon intensity rather than with system demand alone; in the current Korean grid, this implies introducing a midday solar-aligned discount alongside the existing overnight window, rather than relying on the latter exclusively. Second, the workplace-charging infrastructure should be expanded so that solar generation can be absorbed by EV

load during midday hours, complementing rather than competing with overnight residential charging. Third, managed-charging and smart-charger timer systems—of the type evaluated in Powell et al. (2022) and Metcalfe et al. (2026)—should be promoted to reduce the behavioral cost of optimal-hour charging to near zero, since the avoided social damage of T4's emissions under our central assumptions defines a break-even budget of roughly $100 per driver-year for any alternative intervention that achieves the same kWh shift while routing it into low-CI hours. As solar penetration rises, the environmental case for off-peak overnight charging will weaken further, and the policy lever that today produces an inverted abatement cost of approximately −$41 per ton may need to be redesigned around midday hours.

5 Conclusion

This paper uses a randomized controlled trial to estimate the effects of financial incentives on the timing of EV charging in South Korea. We find that a 40% discount on off-peak charging (22:00–08:00), sustained for two months, significantly increases both the probability and volume of off-peak charging—by 150% and 196% of control means, respectively. A 20% discount produces smaller, largely insignificant effects. Neither discount level consistently reduces non-off-peak charging on the platform we observe, suggesting that the primary margin of adjustment is additive rather than substitutive, although platform-only data cannot rule out cross-platform substitution. Translating these intent-to-treat estimates into an arc price elasticity of off-peak energy yields $\sigma \approx -1.45$ in the pooled two-month window, placing the Korean residential setting between the residential off-peak elasticity reported for Calgary (−1.59; Bailey et al. 2025b) and the public-charging elasticity reported for the United Kingdom (−1.45 to −1.00; Bernard et al. 2025), and well above the workplace flat-discount elasticity reported for UCSD (−0.46; Garg et al. 2025).

Several features of the results warrant interpretation. The nonlinear response to incentive magnitude—with 20% producing little effect and 40% generating substantial change—implies that marginal adjustments to existing TOU rate differentials are unlikely to shift charging behavior. This is consistent with the broader demand-response literature, which finds that the peak-to-off-peak price ratio must be sufficiently large to overcome behavioral inertia

(Faruqui and Sergici 2017). The within-duration comparisons further suggest that the elasticity is not realized instantaneously but rises from approximately −0.91 in the one-month arms to approximately −1.99 in the two-month arms, consistent with a habit-formation channel in which drivers internalize the discount only as the off-peak charging routine becomes habitual. The finding that effects are concentrated on slow chargers reinforces that overnight home charging is the most elastic margin, a pattern also observed in Canadian (Bailey et al. 2025b) and UK (Bernard et al. 2025) settings. The heterogeneity by home charger access is informative: participants with home fast chargers increase charging across all time periods when incentivized, while those without them engage in more substitution from non-off-peak to off-peak hours. Read alongside Garg et al. (2025), the comparison suggests that the environmental sign of an EV-charging incentive is jointly determined by instrument design and the dominant charging margin: a flat discount applied to a congested workplace network pulls drivers away from the solar peak in a solar-abundant grid, while a time-conditional discount applied to residential overnight slow charging pulls drivers into hours of higher operating-margin carbon intensity in a coal-backed overnight grid.

Three policy observations follow from our findings. First, as Korea's EV fleet continues to grow, the alignment between charging behavior and grid capacity will become an increasingly important determinant of system costs and environmental outcomes. Our results suggest that financial incentives can shift charging toward off-peak periods, but the magnitude of the price signal matters considerably. Second, the concentration of effects on slow chargers suggests that policies targeting residential charging infrastructure—through smart chargers with timer functionality (Crozier et al. 2020), or through automated scheduling systems like those evaluated by Metcalfe et al. (2026)—may complement price-based interventions by reducing the behavioral cost of off-peak charging to near zero. Third, the environmental and welfare consequences of the discount run in the opposite direction from a conventional climate-policy reading. Under our central assumptions, drawing on Korea Power Exchange (KPX EPSIS) 2024 hourly dispatch data and the Garg-style accounting framework, the T4 discount produces an inverted abatement cost of approximately −\$41 per ton of CO_2 added at a social cost of carbon of \$185/tCO_2; and because the Korean off-peak EV charging tariff already sits approximately at-cost—narrowly above the short-run marginal cost of overnight

coal generation and narrowly below the wholesale system marginal price—a 40% discount on top of this rate pushes the marginal kWh price below social marginal cost, introducing deadweight loss on the additional consumption in addition to the fiscal transfer from KEPCO and Korean taxpayers to participating drivers. Korean policymakers should therefore pair any expansion of off-peak discounts with structural reforms: aligning the discount window with the diurnal cycle of marginal carbon intensity rather than with system demand alone, expanding daytime workplace charging infrastructure to absorb the growing share of solar generation in the Korean grid, and promoting managed-charging or smart-charger timer systems whose welfare consequences may be more favorable under Korean tariff conditions.

Several features of our design temper these conclusions. The sample of 453 participants, while adequate for detecting the large T4 effects, limits statistical power for the smaller effects in the 20% arms and for fine-grained heterogeneity analyses. The study population consists of users of a single commercial charging platform, so the measured increase in off-peak charging cannot be definitively distinguished from substitution across platforms; the absence of a corresponding reduction in non-off-peak charging on the platform is suggestive but not conclusive. The intervention period is relatively short, and our observation window does not yet permit a clean test of whether the behavioral effect persists once the discount is removed—a limitation shared with much of the existing literature, though Bailey et al. (2025b) find that behavior reverts once incentives end. The interaction between EV load shifting and the growing role of solar generation, where daytime charging may become preferable to overnight charging as Korean solar penetration rises, represents a natural avenue for future research (La Nauze et al. 2024).

As we mentioned earlier, our study is not without its limitations. The sample of 453 participants, whereas adequate for detecting the large T4 effects, limits statistical power for detecting smaller effects in the 20% arms and in heterogeneity analyses. The study population consists of users of a single commercial charging platform, and results may not generalize to all EV drivers or to drivers who use multiple platforms. Because we observe only charging sessions conducted through TBU, we cannot rule out that some of the measured increase in off-peak charging reflects substitution from other platforms rather than a net increase in off-peak load—though the absence of a reduction in non-off-peak charging on

TBU itself is suggestive that the response is not purely cross-platform substitution. The intervention period is relatively short (one to two months), and we do not have sufficient post-treatment follow-up to assess the persistence of behavioral changes after incentives are removed—a limitation shared with much of the existing literature, though Bailey et al. (2025b) find that behavior reverts once incentives end. The interaction between EV load shifting and the growing role of solar generation—where daytime charging may be preferable to overnight charging in high-solar-penetration grids—represents an avenue for future research (International Energy Agency 2024; La Nauze et al. 2024).

Acknowledgements

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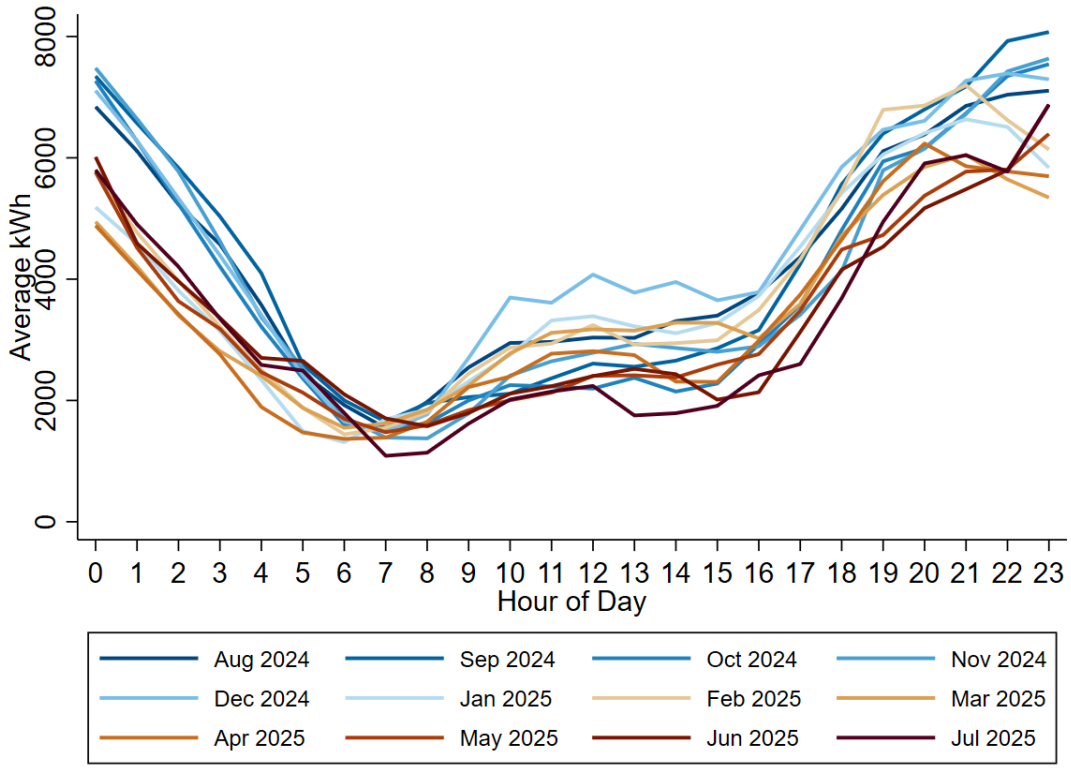
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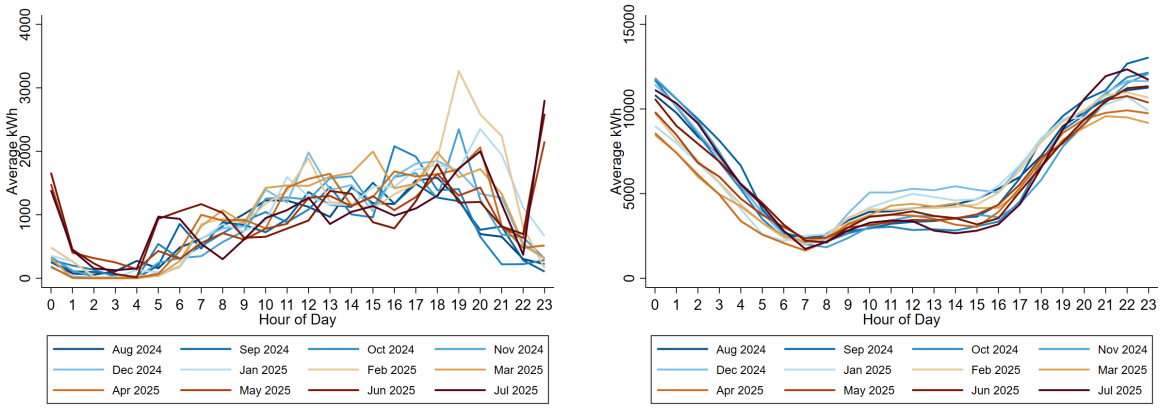
Appendix A

Figure A1: Monthly Pre-Treatment EV Charging Load by Hour



Note: Comparison of EV charging load by month for pre-treatment periods.

Figure A2: Monthly Pre-Treatment EV Charging Load by Hour: Fast and Slow Charger Type

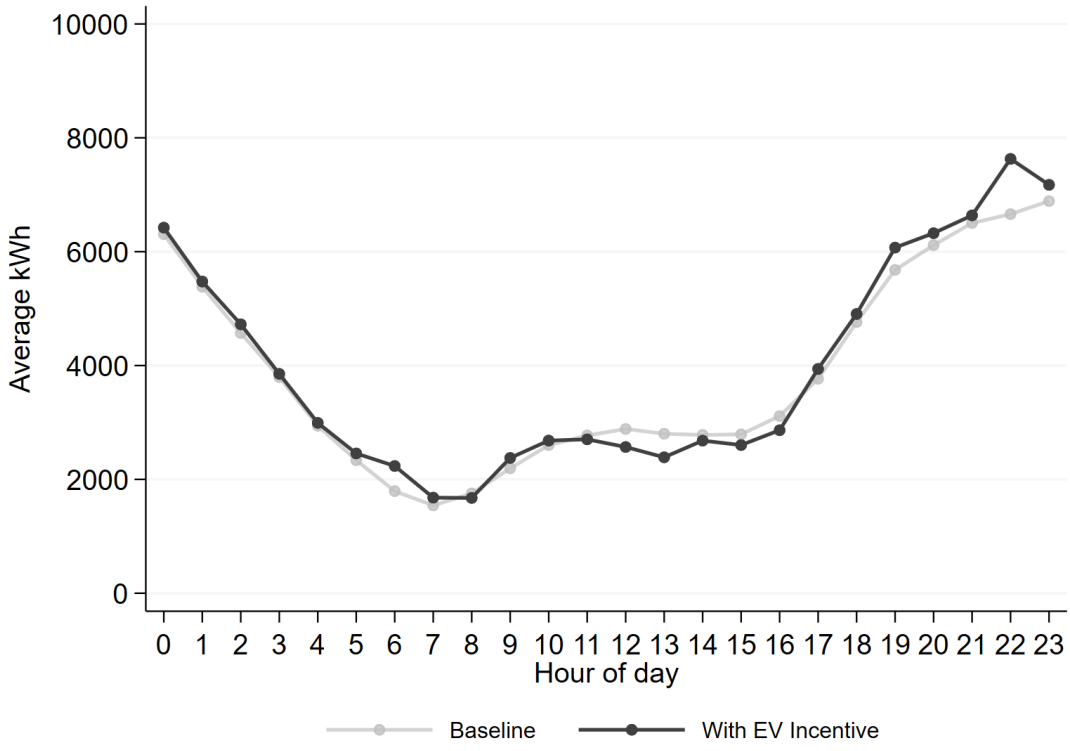


(a) Fast chargers

(b) Slow chargers

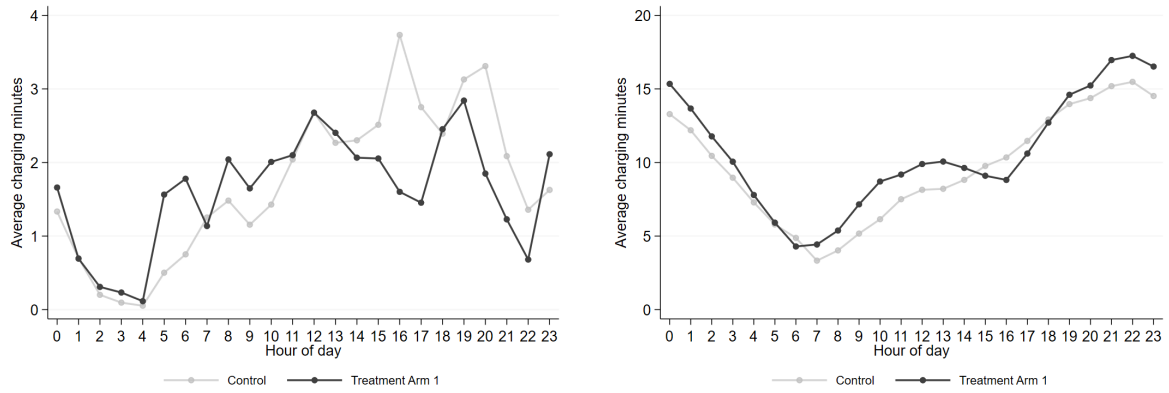
Note: Comparison of EV charging load by month for pre-treatment periods.

Figure A3: EV Charging Load by Hour: Baseline vs. Post-Treatment



Note: Comparison of EV charging load for pre-treatment periods and post-treatment periods for all treatment arms, including the control group.

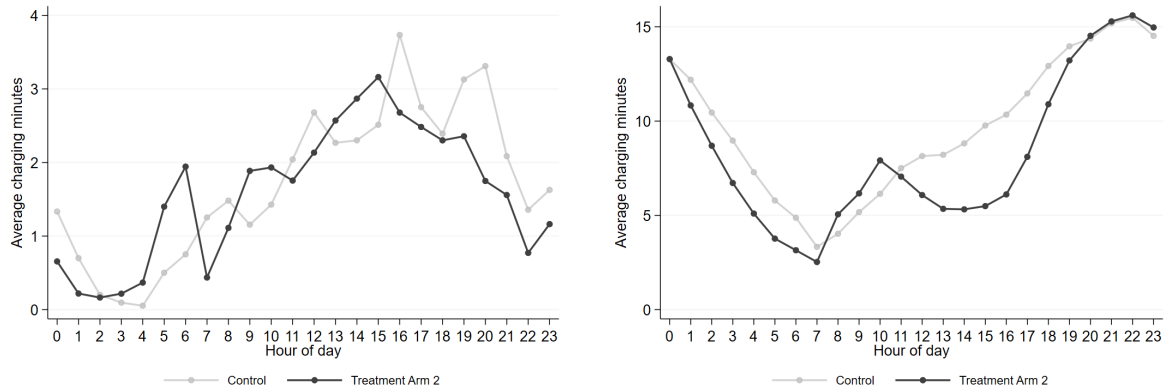
Figure A4: Average Pre-Treatment Charging Minutes: Treatment Arm 1 vs. Control



(a) Fast chargers (b) Slow chargers

Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group.

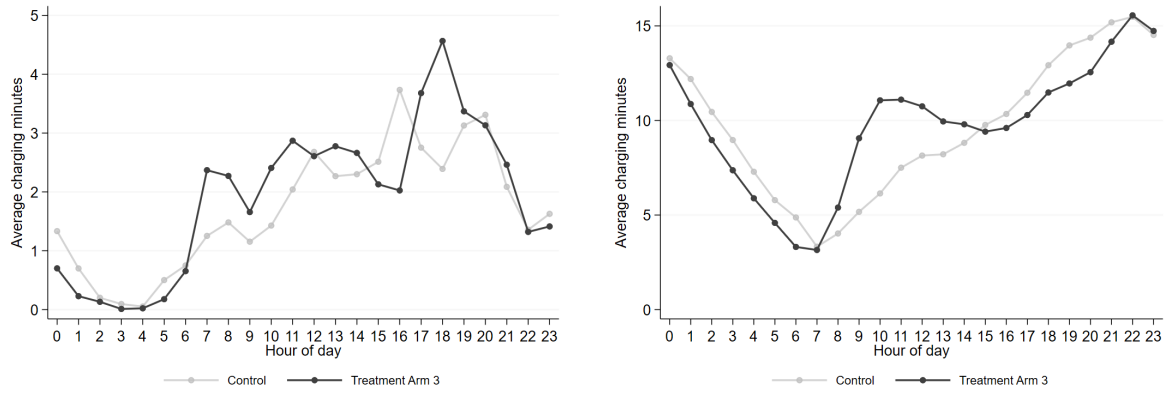
Figure A5: Average Pre-Treatment Charging Minutes: Treatment Arm 2 vs. Control



(a) Fast chargers (b) Slow chargers

Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group.

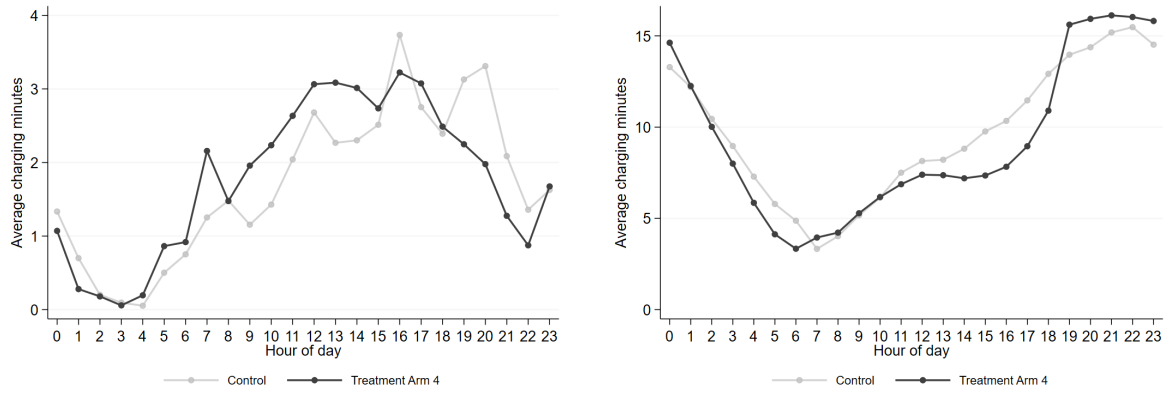
Figure A6: Average Pre-Treatment Charging Minutes: Treatment Arm 3 vs. Control



(a) Fast chargers (b) Slow chargers

Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group.

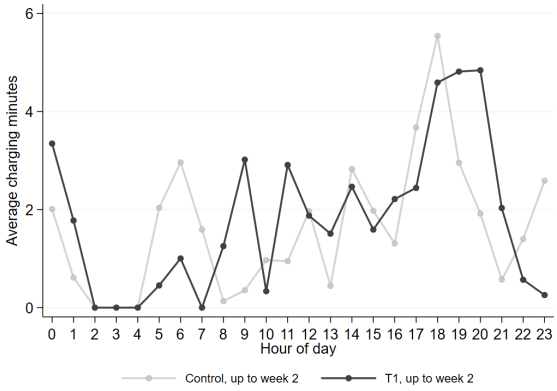
Figure A7: Average Pre-Treatment Charging Minutes: Treatment Arm 4 vs. Control



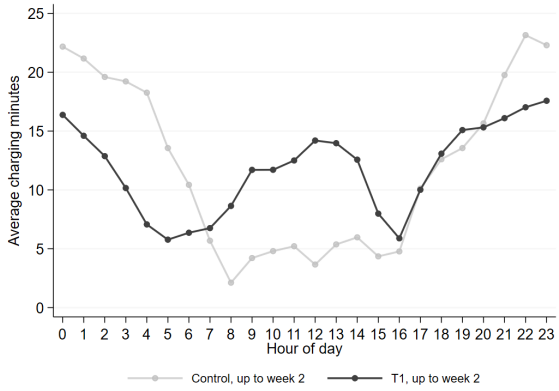
(a) Fast chargers (b) Slow chargers

Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group.

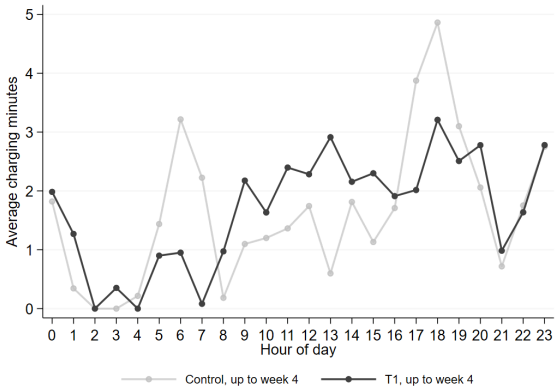
Figure A8: Average Post-Treatment Charging Minutes: Treatment Arm 1 vs. Control



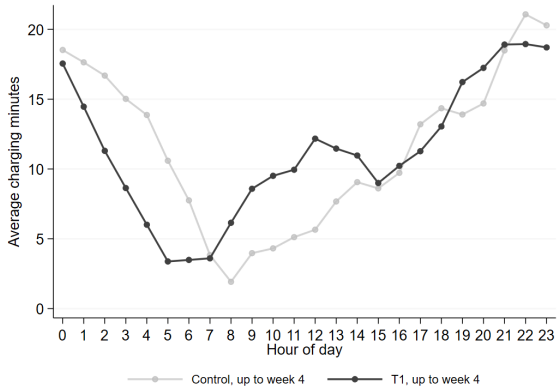
(a) Fast chargers (up to 2 weeks post-treatment)



(b) Slow chargers (up to 2 weeks post-treatment)



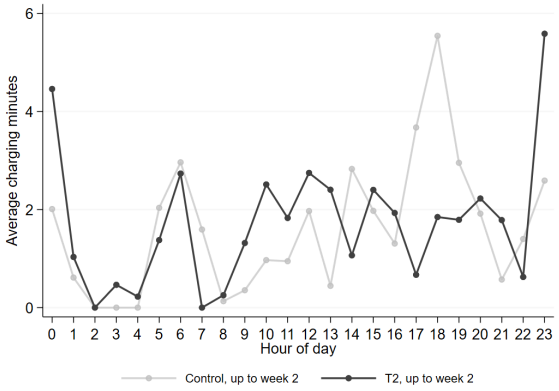
(c) Fast chargers (up to 4 weeks post-treatment)



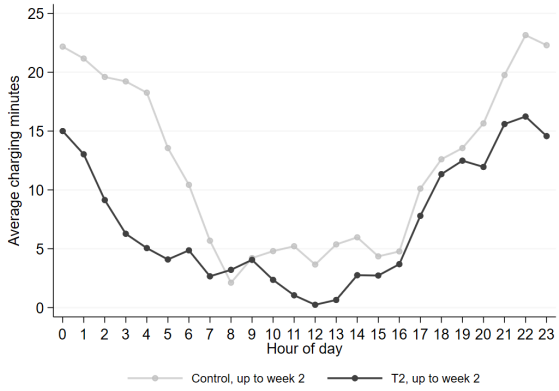
(d) Slow chargers (up to 4 weeks post-treatment)

Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group.

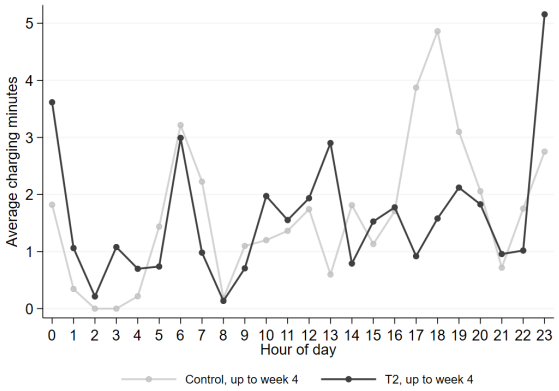
Figure A9: Average Post-Treatment Charging Minutes: Treatment Arm 2 vs. Control



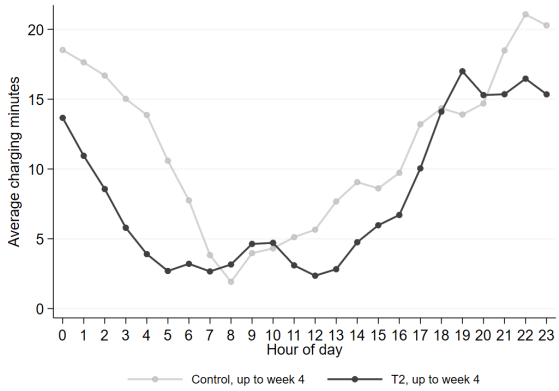
(a) Fast chargers (up to 2 weeks post-treatment)



(b) Slow chargers (up to 2 weeks post-treatment)



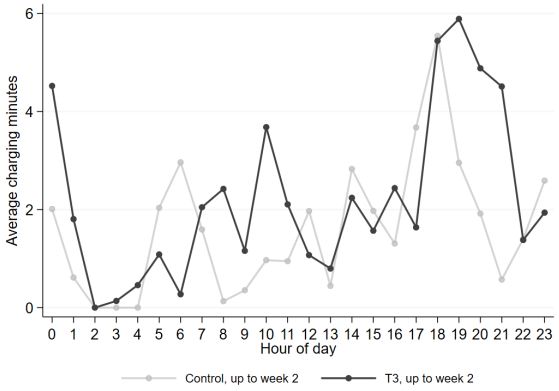
(c) Fast chargers (up to 4 weeks post-treatment)



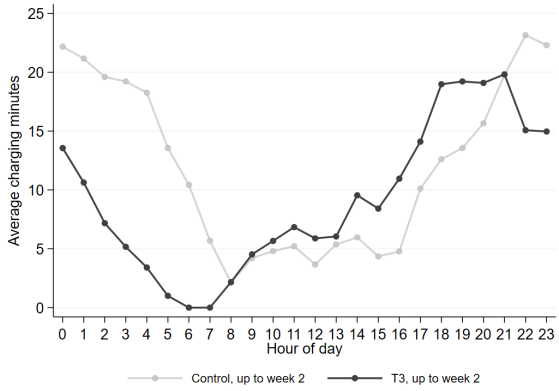
(d) Slow chargers (up to 4 weeks post-treatment)

Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group.

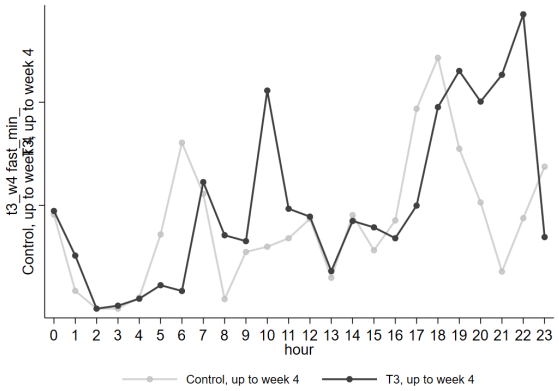
Figure A10: Average Post-Treatment Charging Minutes: Treatment Arm 3 vs. Control



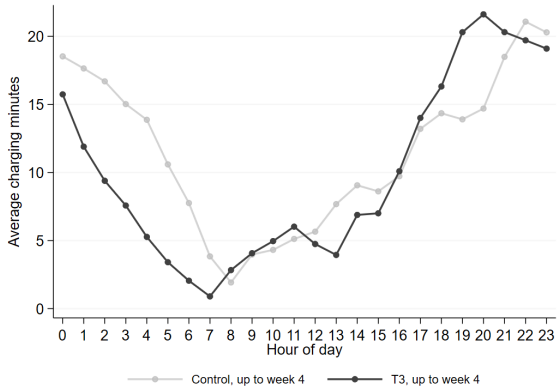
(a) Fast chargers (up to 2 weeks post-treatment)



(b) Slow chargers (up to 2 weeks post-treatment)



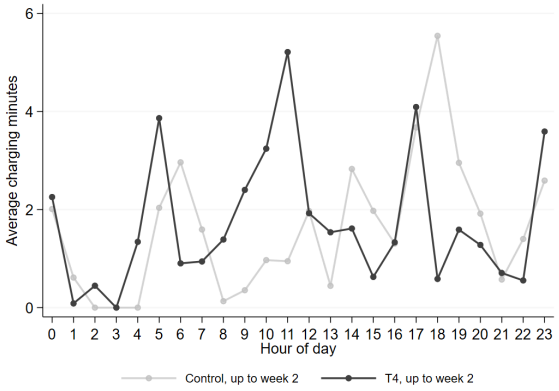
(c) Fast chargers (up to 4 weeks post-treatment)



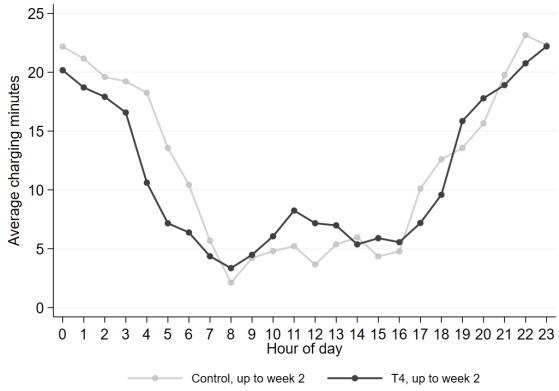
(d) Slow chargers (up to 4 weeks post-treatment)

Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group.

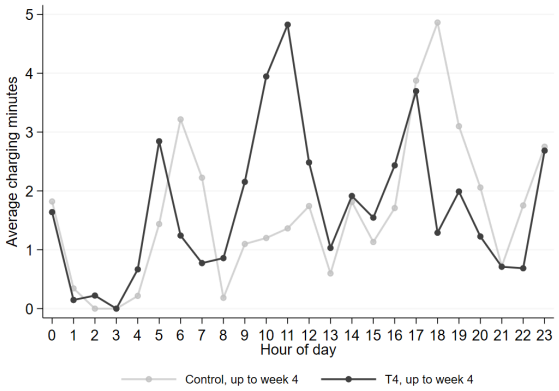
Figure A11: Average Post-Treatment Charging Minutes: Treatment Arm 4 vs. Control



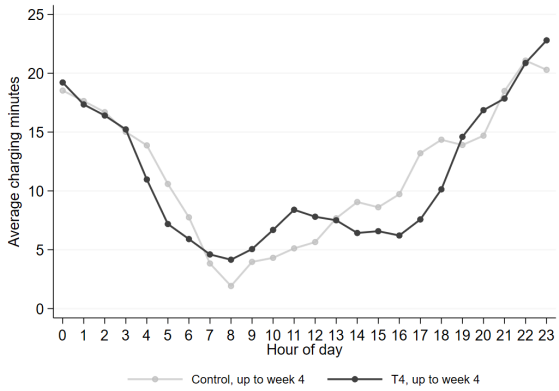
(a) Fast chargers (up to 2 weeks post-treatment)



(b) Slow chargers (up to 2 weeks post-treatment)



(c) Fast chargers (up to 4 weeks post-treatment)



(d) Slow chargers (up to 4 weeks post-treatment)

Note: In each panel, light gray bars correspond to the control group and dark gray bars correspond to the treatment group.

Figure A12: Time Series Total Minutes in TBU, Weekly

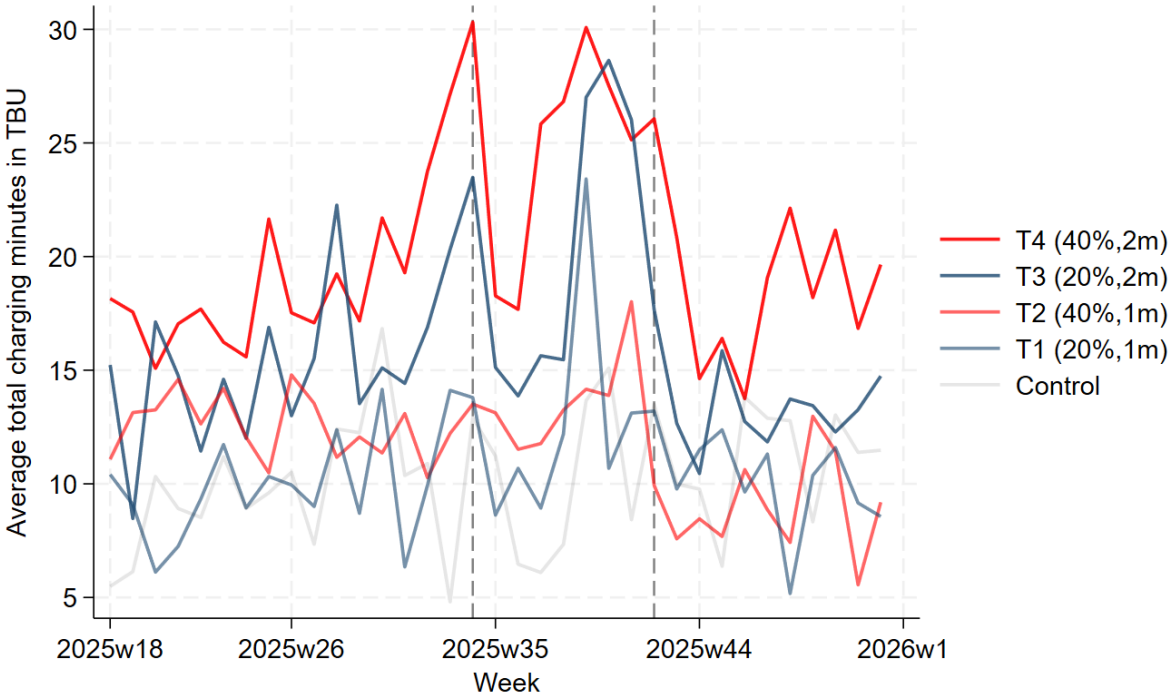


Figure A13: Time Series Total kWh in TBU, Weekly

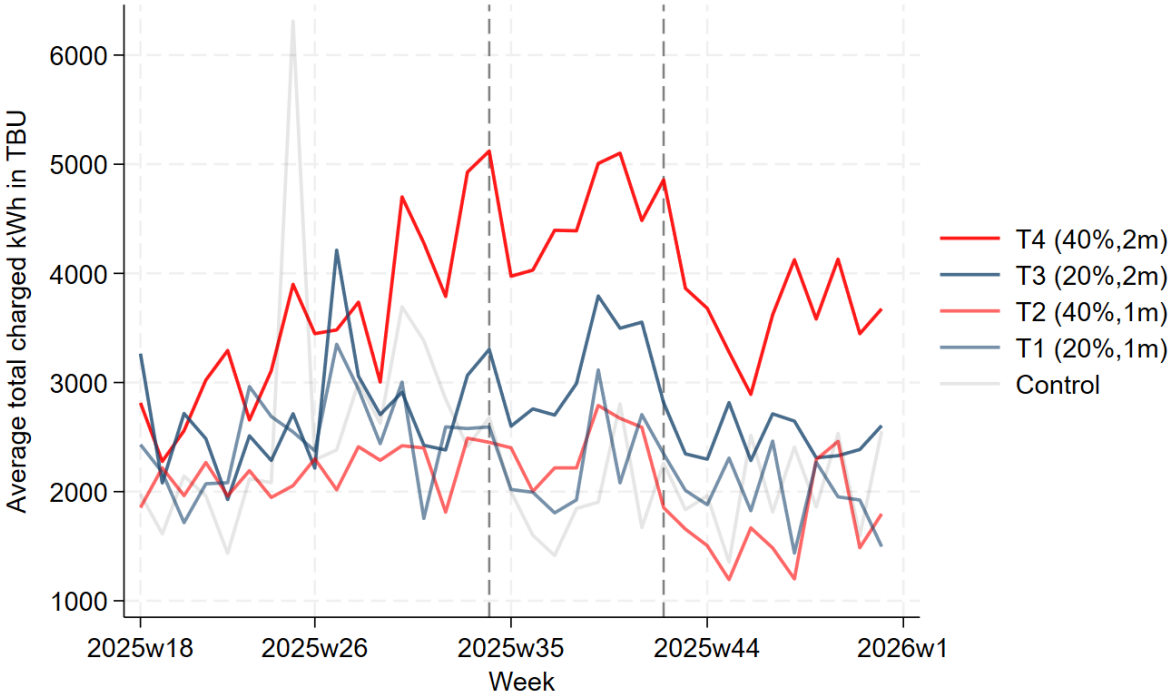
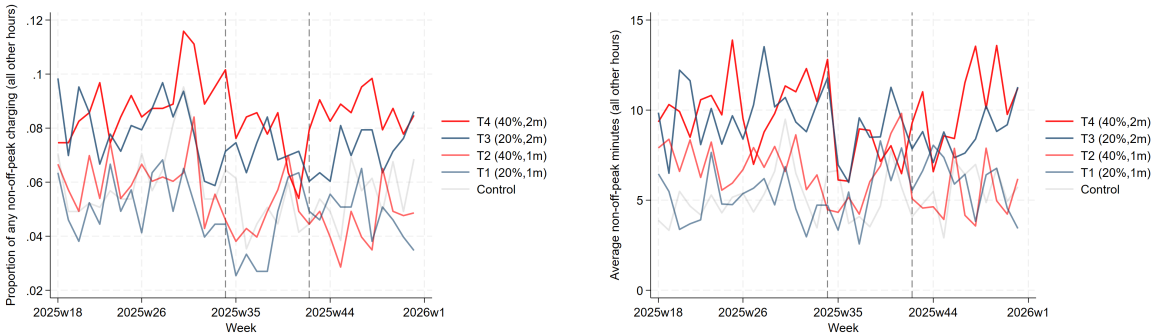
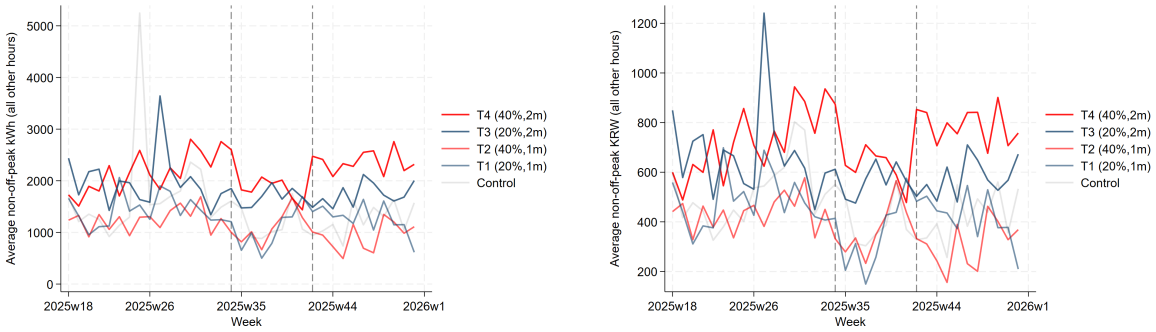


Figure A14: Weekly Non-Off-peak Outcomes (All Other Hours)



(a) Any non-off-peak minutes

(b) Non-off-peak minutes

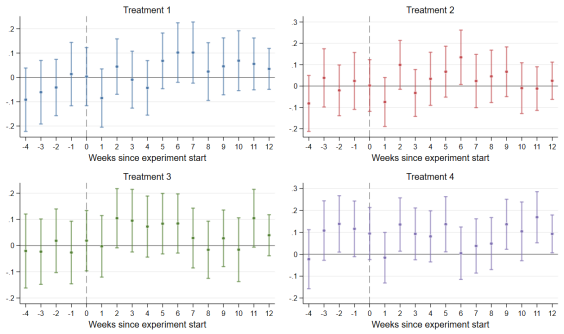


(c) Non-off-peak kWh

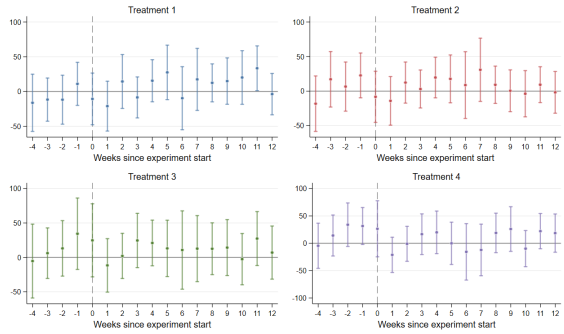
(d) Non-off-peak KRW

Note: Each panel plots weekly averages for treatment (all arms combined) and control groups. The vertical dashed line indicates the start of treatment.

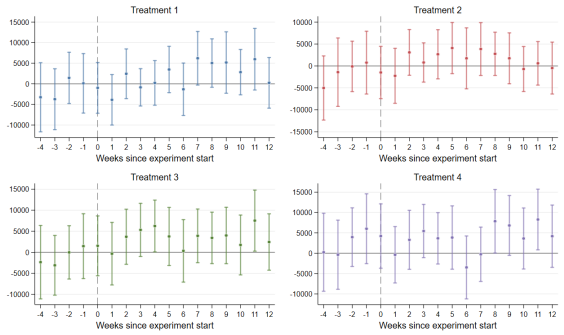
Figure A15: Event-Study Estimates of Non-Off-Peak Charging Outcomes, Weekly



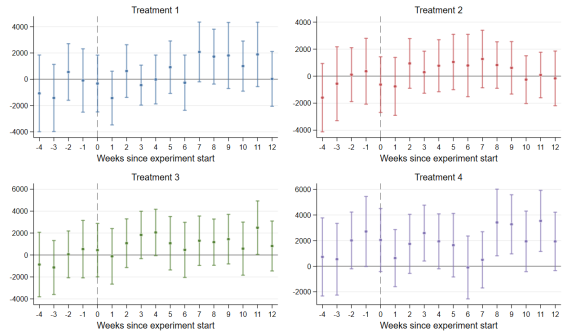
(a) Any non-off-peak charging



(b) Non-off-peak charging minutes



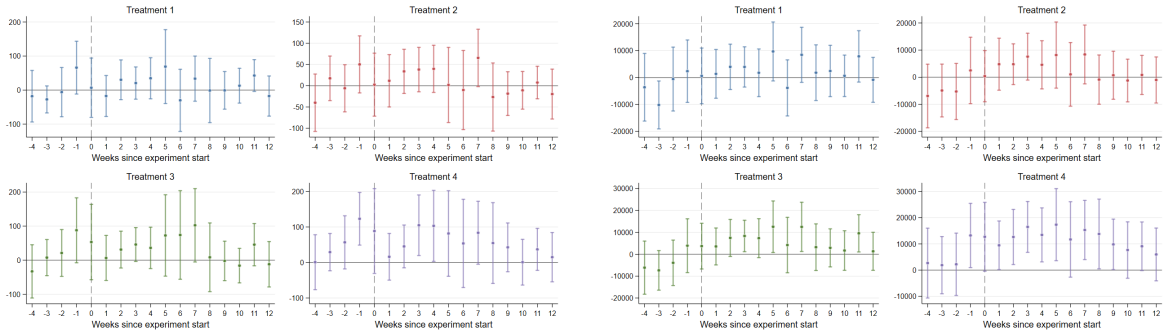
(c) Non-off-peak kWh



(d) Non-off-peak KRW

Note: Each panel plots weekly event-study coefficients from Equation 1, with the week immediately before treatment onset as the reference period. Shaded bands represent 95% confidence intervals based on standard errors clustered at the individual level.

Figure A16: Event-Study Estimates of Total Charging Outcomes in TBU, Weekly

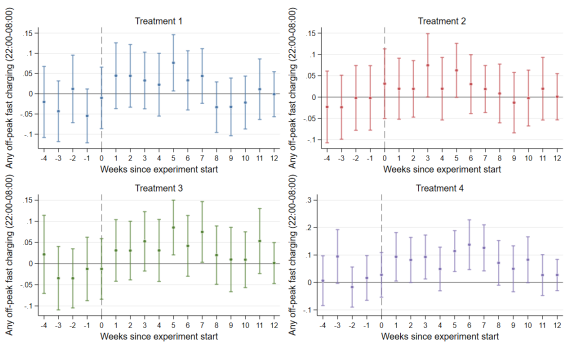


(a) Total charging minutes

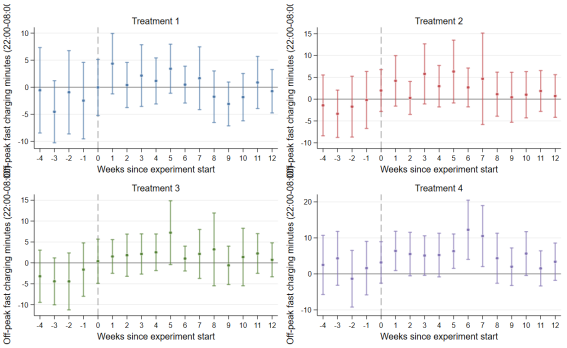
(b) Total kWh

Note: Each panel plots weekly event-study coefficients from Equation 1, with the week immediately before treatment onset as the reference period. Shaded bands represent 95% confidence intervals based on standard errors clustered at the individual level.

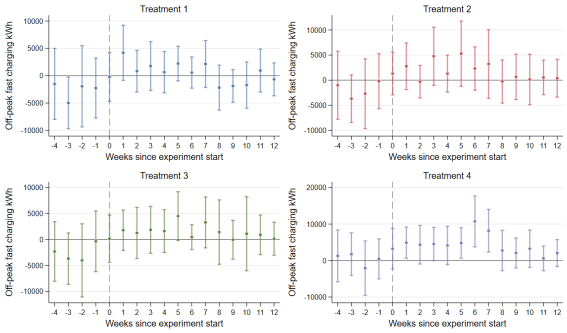
Figure A17: Event-Study Estimates of Off-Peak Charging Outcomes on Fast Chargers, Weekly



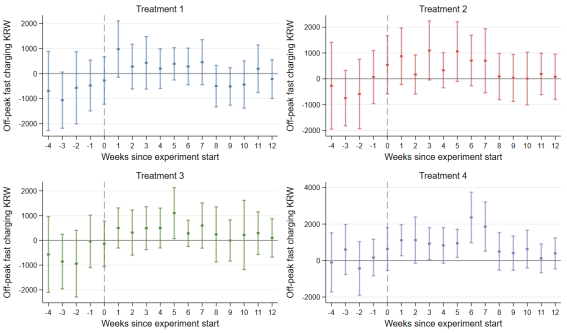
(a) Any off-peak charging



(b) Off-peak charging minutes



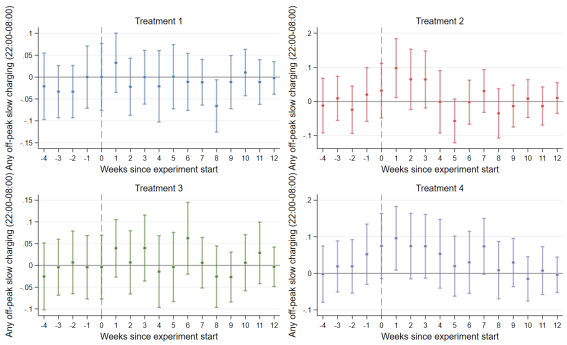
(c) Off-peak kWh



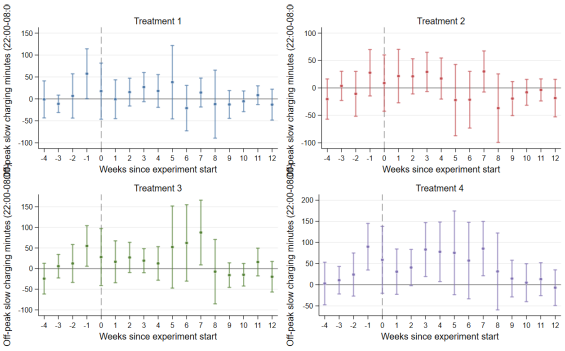
(d) Off-peak KRW

Note: Each panel plots weekly event-study coefficients from Equation 1, with the week immediately before treatment onset as the reference period. Shaded bands represent 95% confidence intervals based on standard errors clustered at the individual level.

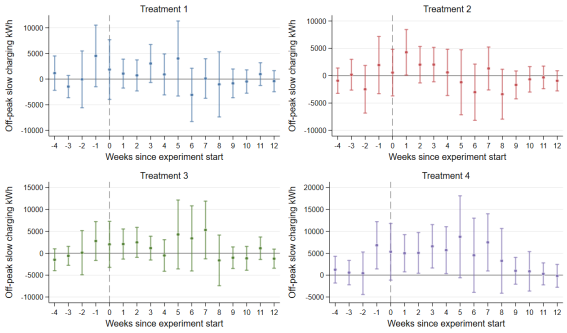
Figure A18: Event-Study Estimates of Off-Peak Charging Outcomes on Slow Chargers, Weekly



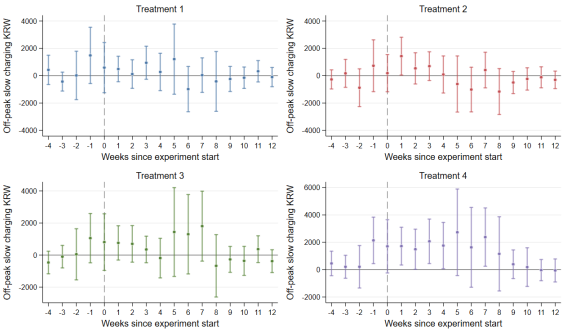
(a) Any off-peak charging



(b) Off-peak charging minutes



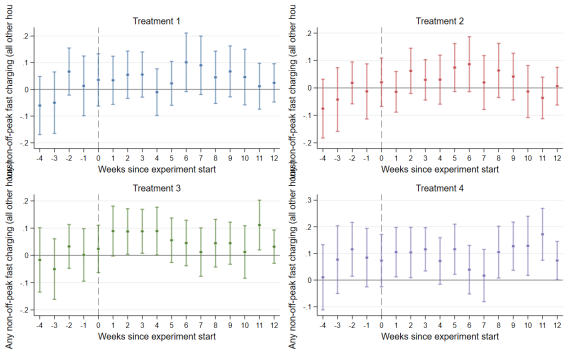
(c) Off-peak kWh



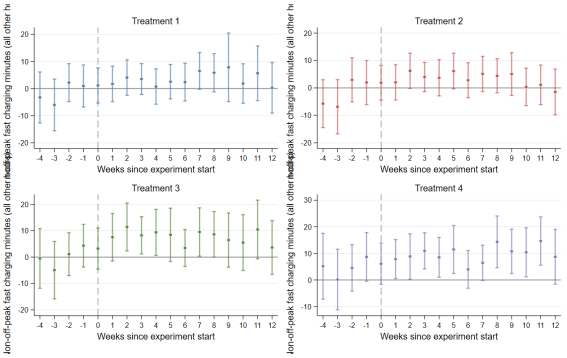
(d) Off-peak KRW

Note: Each panel plots weekly event-study coefficients from Equation 1, with the week immediately before treatment onset as the reference period. Shaded bands represent 95% confidence intervals based on standard errors clustered at the individual level.

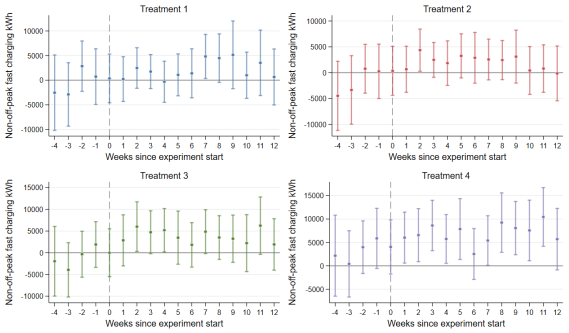
Figure A19: Event-Study Estimates of Non-Off-Peak Charging Outcomes on Fast Chargers, Weekly



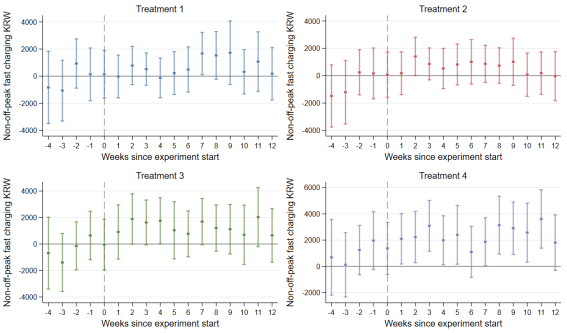
(a) Any non-off-peak charging



(b) Non-off-peak charging minutes



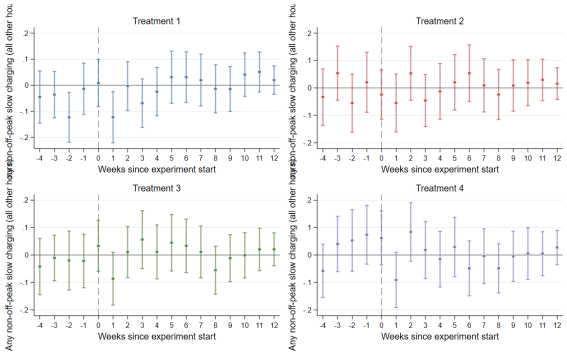
(c) Non-off-peak kWh



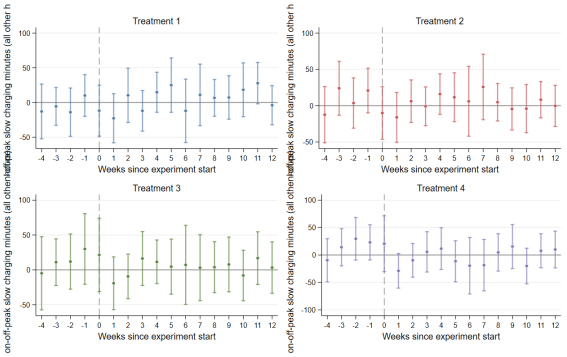
(d) Non-off-peak KRW

Note: Each panel plots weekly event-study coefficients from Equation 1, with the week immediately before treatment onset as the reference period. Shaded bands represent 95% confidence intervals based on standard errors clustered at the individual level.

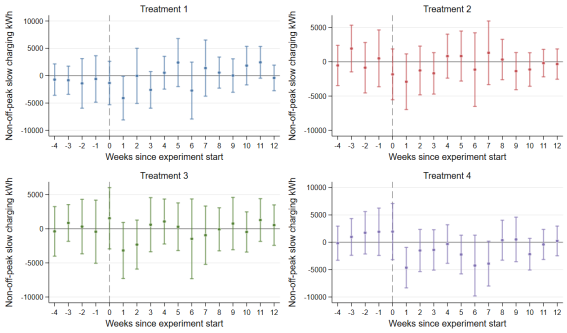
Figure A20: Event-Study Estimates of Non-Off-Peak Charging Outcomes on Slow Chargers, Weekly



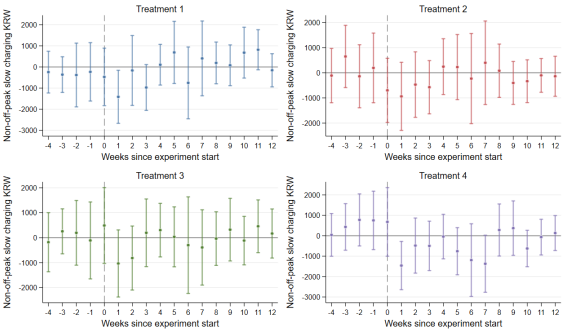
(a) Any non-off-peak charging



(b) Non-off-peak charging minutes



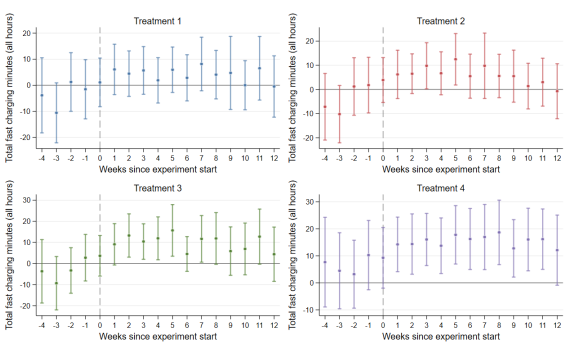
(c) Non-off-peak kWh



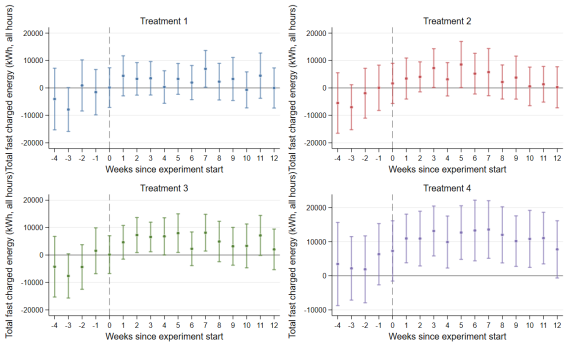
(d) Non-off-peak KRW

Note: Each panel plots weekly event-study coefficients from Equation 1, with the week immediately before treatment onset as the reference period. Shaded bands represent 95% confidence intervals based on standard errors clustered at the individual level.

Figure A21: Event-Study Estimates of Total Charging Outcomes on Fast Chargers in TBU, Weekly



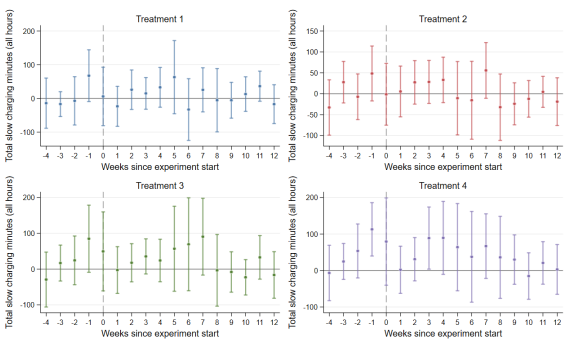
(a) Total charging minutes



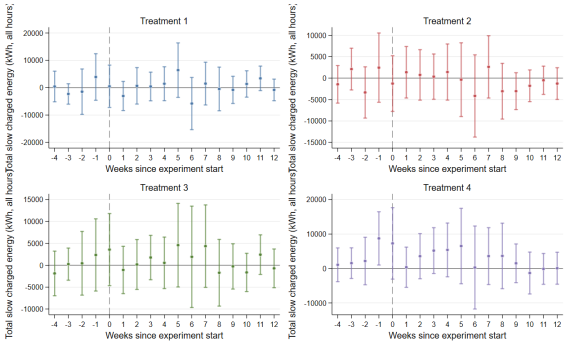
(b) Total kWh

Note: Each panel plots weekly event-study coefficients from Equation 1, with the week immediately before treatment onset as the reference period. Shaded bands represent 95% confidence intervals based on standard errors clustered at the individual level.

Figure A22: Event-Study Estimates of Total Charging Outcomes on Slow Chargers in TBU, Weekly



(a) Total charging minutes



(b) Total kWh

Note: Each panel plots weekly event-study coefficients from Equation 1, with the week immediately before treatment onset as the reference period. Shaded bands represent 95% confidence intervals based on standard errors clustered at the individual level.

Table A1: Treatment Effects by Age Group: Up to First Month Post-Treatment

	Off-peak hours				Non-off-peak hours				Total	
	(1) =1 if any	(2) min.	(3) kWh	(4) KRW	(5) =1 if any	(6) min.	(7) kWh	(8) KRW	(9) min.	(10) kWh
Age: 20–39										
T1 × Post	0.03**	3.46	993.54**	258.35**	0.02	2.24	255.97	63.17	5.70	1249.51*
	(0.01)	(2.60)	(480.69)	(120.43)	(0.02)	(4.11)	(510.51)	(175.75)	(5.62)	(754.98)
T2 × Post	0.02*	5.05	548.01	186.10	0.01	1.72	353.55	114.19	6.77	901.56
	(0.01)	(4.04)	(395.06)	(122.22)	(0.02)	(3.56)	(461.64)	(166.73)	(5.95)	(690.28)
T3 × Post	0.02	-1.20	364.20	68.59	0.04*	5.05	870.80*	271.24	3.86	1235.00
	(0.01)	(2.28)	(529.43)	(122.30)	(0.02)	(3.76)	(493.73)	(173.27)	(4.64)	(845.44)
T4 × Post	0.04**	2.83	1363.41**	330.69**	0.02	-1.37	-385.36	106.12	1.46	978.06
	(0.02)	(2.78)	(564.18)	(137.13)	(0.02)	(3.52)	(592.51)	(160.09)	(5.33)	(932.21)
Control mean	0.03	6.41	697.45	208.12	0.07	9.05	1,514.42	515.12	15.46	2,211.88
Control SD	0.18	54.08	4,871.61	1,450.39	0.26	55.27	6,729.67	2,317.19	89.76	9,135.69
Observations	70,751	70,751	70,751	70,751	70,751	70,751	70,751	70,751	70,751	70,751
Age: 40–59										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
T1 × Post	0.00	1.36	203.89	70.37	-0.02	-1.74	-240.00	-131.82	-0.38	-36.11
	(0.01)	(2.15)	(331.42)	(94.47)	(0.01)	(2.20)	(390.60)	(127.81)	(3.29)	(537.03)
T2 × Post	0.02*	1.85	783.24*	215.44*	0.00	-0.64	196.07	28.58	1.21	979.31
	(0.01)	(1.97)	(419.29)	(109.95)	(0.01)	(1.72)	(344.61)	(116.53)	(3.16)	(638.97)
T3 × Post	0.01	3.28	434.42	139.92*	0.01	-0.68	315.39	74.79	2.60	749.81
	(0.01)	(2.30)	(353.92)	(81.93)	(0.02)	(2.03)	(474.12)	(159.41)	(3.10)	(494.22)
T4 × Post	0.03**	8.71**	1162.70***	347.18***	0.02	0.22	582.21	168.12	8.94*	1744.91***
	(0.01)	(3.66)	(395.01)	(104.46)	(0.02)	(2.19)	(428.09)	(137.26)	(5.24)	(620.65)
Control mean	0.02	2.49	439.48	127.61	0.06	4.77	1,456.48	463.85	7.26	1,895.96
Control SD	0.13	29.18	4,351.64	1,218.57	0.23	28.60	17,586.03	2,321.56	47.99	18,331.24
Observations	144,556	144,556	144,556	144,556	144,556	144,556	144,556	144,556	144,556	144,556
Age: 60+										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
T1 × Post	-0.02	-0.96	-429.38	-136.98	-0.02	-3.70**	-506.26	-193.57	-4.66**	-935.63
	(0.02)	(0.84)	(420.84)	(137.35)	(0.03)	(1.81)	(977.87)	(357.95)	(2.19)	(1254.97)
T2 × Post	-0.02	7.66	276.31	90.27	-0.12	-15.13*	-3680.75	-1261.55	-7.47	-3404.44
	(0.03)	(7.35)	(933.19)	(304.03)	(0.08)	(7.78)	(2643.40)	(907.90)	(12.31)	(3264.69)
T3 × Post	-0.01	-1.91	-541.48	-172.71	-0.02	-19.94	-1557.12	-569.15	-21.85	-2098.60
	(0.03)	(1.17)	(448.83)	(146.96)	(0.09)	(15.81)	(2003.75)	(704.56)	(16.12)	(2201.38)
T4 × Post	0.02	11.57	1245.42	412.22	0.05	-8.22	1165.02	494.01	3.34	2410.45
	(0.03)	(10.43)	(1083.77)	(402.63)	(0.05)	(6.59)	(1748.34)	(694.88)	(4.85)	(1847.36)
Control mean	0.02	1.02	615.84	203.06	0.07	4.60	2,235.02	765.40	5.62	2,850.86
Control SD	0.14	10.29	4,603.96	1,521.70	0.26	23.50	9,191.18	3,175.26	27.94	10,626.82
Observations	14,761	14,761	14,761	14,761	14,761	14,761	14,761	14,761	14,761	14,761

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Treatment Effects by Age Group: Up to Two Months Post-Treatment

	Off-peak hours				Non-off-peak hours				Total	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	=1 if any	min.	kWh	KRW	=1 if any	min.	kWh	KRW	min.	kWh
Age: 20–39										
T1 × Post	0.03**	2.12	736.20	161.92	0.03	1.58	364.60	106.64	3.71	1100.80
	(0.01)	(4.03)	(524.99)	(162.53)	(0.02)	(4.09)	(447.02)	(160.58)	(7.38)	(825.59)
T2 × Post	0.01	2.28	200.27	58.23	0.03	4.44	696.43	226.94	6.72	896.70
	(0.01)	(4.37)	(404.77)	(148.12)	(0.02)	(4.07)	(480.10)	(177.24)	(7.57)	(801.72)
T3 × Post	0.02	0.53	634.26	115.64	0.04*	6.62	794.79	244.81	7.15	1429.05
	(0.01)	(4.93)	(649.80)	(183.76)	(0.02)	(5.00)	(524.61)	(187.24)	(9.04)	(1017.00)
T4 × Post	0.04**	4.24	2044.97**	452.75**	0.00	-3.32	-686.30	0.12	0.92	1358.66
	(0.02)	(4.68)	(839.50)	(214.40)	(0.02)	(4.01)	(605.51)	(159.51)	(7.45)	(1192.12)
Control mean	0.03	6.41	697.45	208.12	0.07	9.05	1,514.42	515.12	15.46	2,211.88
Control SD	0.18	54.08	4,871.61	1,450.39	0.26	55.27	6,729.67	2,317.19	89.76	9,135.69
Observations	74,643	74,643	74,643	74,643	74,643	74,643	74,643	74,643	74,643	74,643
Age: 40–59										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
T1 × Post	0.00	2.86	305.24	107.16	0.00	0.68	153.62	11.11	3.54	458.86
	(0.01)	(2.35)	(293.55)	(87.48)	(0.01)	(1.93)	(379.81)	(125.08)	(3.50)	(527.91)
T2 × Post	0.02*	1.15	732.81*	183.07*	0.01	0.35	385.13	91.74	1.50	1117.94*
	(0.01)	(1.56)	(433.97)	(93.90)	(0.01)	(1.64)	(319.60)	(107.84)	(2.56)	(630.26)
T3 × Post	0.02	8.50**	875.81**	282.63**	0.01	-0.07	401.48	115.33	8.43*	1277.30**
	(0.01)	(4.07)	(415.05)	(110.21)	(0.01)	(1.97)	(446.46)	(150.86)	(4.69)	(528.80)
T4 × Post	0.03***	11.10**	1423.23***	414.94***	0.02	1.16	640.68	186.05	12.26*	2063.91***
	(0.01)	(4.39)	(395.74)	(107.05)	(0.01)	(2.52)	(411.37)	(129.19)	(6.31)	(602.43)
Control mean	0.02	2.49	439.48	127.61	0.06	4.77	1,456.48	463.85	7.26	1,895.96
Control SD	0.13	29.18	4,351.64	1,218.57	0.23	28.60	17,586.03	2,321.56	47.99	18,331.24
Observations	152,508	152,508	152,508	152,508	152,508	152,508	152,508	152,508	152,508	152,508
Age: 60+										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
T1 × Post	-0.01	-2.94	-536.95	-180.85	-0.01	-2.78**	-195.72	-91.21	-5.72*	-732.67
	(0.02)	(2.26)	(496.08)	(163.57)	(0.03)	(1.33)	(834.06)	(298.44)	(3.15)	(1075.95)
T2 × Post	-0.03	0.96	-431.69	-147.94	-0.12	-14.25*	-3404.05	-1169.14	-13.30	-3835.74
	(0.02)	(4.28)	(653.62)	(214.60)	(0.08)	(7.68)	(2592.52)	(885.79)	(10.02)	(2979.10)
T3 × Post	-0.02	-4.16*	-841.97	-279.62	-0.03	-18.82	-1301.20	-485.17	-22.98	-2143.17
	(0.02)	(2.45)	(535.85)	(177.76)	(0.09)	(15.83)	(1927.75)	(672.31)	(16.35)	(2063.95)
T4 × Post	0.03	17.84	1807.95	682.08	0.06	-3.74	1547.51	628.37	14.10	3355.46
	(0.03)	(18.68)	(1788.38)	(713.19)	(0.05)	(3.72)	(1553.87)	(630.46)	(15.49)	(2114.87)
Control mean	0.02	1.02	615.84	203.06	0.07	4.60	2,235.02	765.40	5.62	2,850.86
Control SD	0.14	10.29	4,603.96	1,521.70	0.26	23.50	9,191.18	3,175.26	27.94	10,626.82
Observations	15,573	15,573	15,573	15,573	15,573	15,573	15,573	15,573	15,573	15,573

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Treatment Effects by Home Charger Access: Up to Two Months Post-Treatment

	Off-peak hours				Non-off-peak hours				Total	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	=1 if any	min.	kWh	KRW	=1 if any	min.	kWh	KRW	min.	kWh
Home Charger Access: = 1 if any fast charger										
T1 × Post	0.02	3.67*	453.19	158.79*	0.04***	5.42**	1038.12***	296.97***	9.09***	1491.31***
	(0.01)	(2.13)	(301.36)	(91.48)	(0.01)	(2.41)	(352.16)	(111.42)	(3.46)	(502.34)
T2 × Post	0.01	1.59	334.07	104.99	0.02	1.83	606.34	157.56	3.41	940.41*
	(0.01)	(2.19)	(283.79)	(81.89)	(0.02)	(2.16)	(408.59)	(138.51)	(3.36)	(559.06)
T3 × Post	0.01	1.28	820.12	188.63	0.03**	3.03*	813.86	242.05	4.31	1633.98**
	(0.01)	(2.76)	(547.13)	(119.87)	(0.02)	(1.71)	(566.34)	(193.11)	(3.43)	(641.01)
T4 × Post	0.04***	9.53**	1894.90***	530.09***	0.04**	3.31	874.24	378.05**	12.84*	2769.14***
	(0.01)	(4.58)	(558.80)	(140.71)	(0.02)	(3.03)	(558.07)	(176.10)	(6.84)	(826.24)
Pre-Treat Control: mean	0.02	3.41	526.64	147.40	0.07	6.21	1,553.37	489.26	9.62	2,080.01
Pre-Treat Control: SD	0.15	38.18	4,495.69	1,194.55	0.25	39.56	18,069.84	2,349.92	63.46	18,820.56
Observations	123,510	123,510	123,510	123,510	123,510	123,510	123,510	123,510	123,510	123,510
Home Charger Access: no fast charger										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
T1 × Post	-0.00	-0.12	212.29	12.34	-0.03**	-5.15*	-871.49**	-300.02*	-5.27	-659.20
	(0.01)	(3.70)	(431.52)	(136.92)	(0.02)	(2.78)	(437.40)	(160.04)	(6.12)	(737.45)
T2 × Post	0.02	0.93	603.81	120.88	-0.00	-0.37	-71.33	-27.57	0.56	532.48
	(0.01)	(3.19)	(544.51)	(141.40)	(0.01)	(2.90)	(374.92)	(140.57)	(5.69)	(821.08)
T3 × Post	0.02	8.30	540.73	181.07	-0.00	-2.33	-165.69	-61.22	5.97	375.04
	(0.01)	(5.50)	(450.49)	(153.88)	(0.02)	(3.95)	(433.10)	(156.40)	(7.89)	(729.29)
T4 × Post	0.02*	9.25*	1404.80**	369.27*	-0.01	-5.41*	-377.06	-75.07	3.84	1027.75
	(0.01)	(5.58)	(578.39)	(195.65)	(0.01)	(2.93)	(432.25)	(144.31)	(7.07)	(788.85)
Pre-Treat Control: mean	0.02	3.96	548.32	174.97	0.06	6.04	1,507.81	521.40	9.99	2,056.13
Pre-Treat Control: SD	0.15	38.50	4,612.24	1,474.47	0.24	38.31	7,012.72	2,456.28	64.05	9,145.06
Observations	119,214	119,214	119,214	119,214	119,214	119,214	119,214	119,214	119,214	119,214

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Treatment Effects by Workplace Charger Access: Up to Two Months Post-Treatment

	Off-peak hours				Non-off-peak hours				Total	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	=1 if any	min.	kWh	KRW	=1 if any	min.	kWh	KRW	min.	kWh
Workplace Charger Access: = 1 if any fast charger										
T1 × Post	0.01	-1.41	136.51	0.14	0.02	-0.09	211.12	31.02	-1.50	347.63
	(0.01)	(2.58)	(345.49)	(116.65)	(0.02)	(3.54)	(520.95)	(181.52)	(5.65)	(722.56)
T2 × Post	-0.00	-1.21	-128.37	-54.07	-0.00	-2.06	-26.94	-21.71	-3.26	-155.30
	(0.01)	(2.63)	(297.21)	(109.08)	(0.02)	(3.27)	(541.28)	(195.35)	(5.48)	(747.55)
T3 × Post	0.01	4.94	565.47	152.34	0.02	-3.72	151.32	42.56	1.22	716.79
	(0.01)	(6.03)	(604.49)	(189.87)	(0.02)	(3.74)	(538.74)	(192.27)	(8.14)	(927.77)
T4 × Post	0.04***	11.29**	2314.89***	592.53***	0.02	-1.54	478.90	151.31	9.74	2793.79***
	(0.01)	(5.65)	(670.58)	(183.58)	(0.02)	(3.98)	(580.38)	(193.04)	(8.87)	(964.76)
Pre-Treat Control: mean	0.03	3.34	633.87	198.89	0.07	7.04	1,761.52	603.43	10.38	2,395.39
Pre-Treat Control: SD	0.16	32.09	4,816.82	1,507.19	0.26	40.86	7,782.01	2,687.06	60.91	9,853.29
Observations	98,271	98,271	98,271	98,271	98,271	98,271	98,271	98,271	98,271	98,271
Workplace Charger Access: no fast charger										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
T1 × Post	0.00	4.62	509.10	166.88	-0.00	1.42	171.08	30.79	6.05	680.18
	(0.01)	(2.91)	(360.74)	(104.12)	(0.01)	(1.71)	(300.67)	(96.23)	(3.84)	(528.98)
T2 × Post	0.03**	3.32	902.06**	242.03**	0.02	3.60*	602.01**	164.30*	6.92**	1504.07**
	(0.01)	(2.39)	(458.01)	(102.64)	(0.01)	(1.85)	(271.82)	(87.47)	(3.33)	(618.99)
T3 × Post	0.02	5.81	753.30*	225.68**	0.02	3.40	504.77	139.75	9.21*	1258.07**
	(0.01)	(3.60)	(405.21)	(98.96)	(0.01)	(2.56)	(420.39)	(139.61)	(4.70)	(500.02)
T4 × Post	0.02**	8.39*	1179.58**	357.04**	0.02	0.15	218.77	192.41	8.55*	1398.35**
	(0.01)	(4.53)	(471.01)	(151.23)	(0.01)	(1.97)	(434.88)	(130.85)	(4.94)	(656.44)
Pre-Treat Control: mean	0.02	3.90	455.34	126.62	0.06	5.39	1,346.21	420.45	9.29	1,801.56
Pre-Treat Control: SD	0.14	42.77	4,308.92	1,146.87	0.23	37.44	18,109.27	2,123.67	65.94	18,803.87
Observations	144,453	144,453	144,453	144,453	144,453	144,453	144,453	144,453	144,453	144,453

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Treatment Effects by Pre-Treatment Charger-Type Usage Frequency: Up to Two Months Post-Treatment

	Fast chargers				Slow chargers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	=1 if any	min. rate	kWh rate	KRW rate	=1 if any	min. rate	kWh rate	KRW rate
Panel A. High Fast Session Rate Users								
T1 × Post	0.019	0.018	0.018	0.018	-0.005	-0.005	-0.005	-0.005
	(0.014)	(0.014)	(0.014)	(0.014)	(0.005)	(0.005)	(0.006)	(0.006)
T2 × Post	0.031**	0.031**	0.031**	0.031**	-0.003	-0.004	-0.004	-0.004
	(0.014)	(0.014)	(0.014)	(0.014)	(0.006)	(0.006)	(0.006)	(0.006)
T3 × Post	0.025**	0.024**	0.025**	0.025**	-0.006	-0.006	-0.006	-0.006
	(0.011)	(0.011)	(0.011)	(0.011)	(0.006)	(0.005)	(0.005)	(0.005)
T4 × Post	0.046***	0.046***	0.046***	0.046***	-0.000	-0.001	-0.001	-0.001
	(0.013)	(0.013)	(0.013)	(0.013)	(0.006)	(0.006)	(0.006)	(0.006)
Pre-treat Control: mean	0.022	0.022	0.022	0.022	0.003	0.003	0.003	0.003
Pre-treat Control: SD	0.148	0.147	0.147	0.147	0.057	0.052	0.051	0.051
Observations	94,512	94,512	94,512	94,512	94,512	94,512	94,512	94,512
Panel B. High Slow Session Rate Users								
T1 × Post	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.013)	(0.013)	(0.013)	(0.013)
T2 × Post	0.002**	0.002*	0.002*	0.002*	-0.000	-0.000	-0.000	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.015)	(0.015)	(0.015)	(0.015)
T3 × Post	0.004*	0.003	0.003	0.003	0.015	0.015	0.014	0.015
	(0.002)	(0.002)	(0.002)	(0.002)	(0.017)	(0.017)	(0.017)	(0.017)
T4 × Post	0.009*	0.008*	0.008*	0.008*	0.017	0.017	0.017	0.017
	(0.005)	(0.005)	(0.005)	(0.005)	(0.014)	(0.014)	(0.014)	(0.014)
Pre-treat Control: mean	0.002	0.002	0.002	0.002	0.036	0.036	0.036	0.036
Pre-treat Control: SD	0.044	0.042	0.042	0.042	0.185	0.185	0.185	0.185
Observations	94,512	94,512	94,512	94,512	94,512	94,512	94,512	94,512

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Treatment Effects by Day of Week: Up to First Month Post-Treatment

	Off-peak hours				Non-off-peak hours			
	(1) =1 if any	(2) min.	(3) kWh	(4) KRW	(5) =1 if any	(6) min.	(7) kWh	(8) KRW
Panel A: Weekday								
T1 × Post	0.01	1.60	547.96*	147.45*	-0.01	-0.83	-175.64	-89.64
	(0.01)	(2.01)	(289.04)	(81.08)	(0.01)	(1.74)	(312.75)	(104.88)
T2 × Post	0.03***	2.76	824.00***	238.13***	-0.00	-0.68	72.71	0.05
	(0.01)	(2.05)	(309.22)	(89.98)	(0.01)	(1.42)	(280.38)	(95.57)
T3 × Post	0.01	0.45	376.98	87.48	0.01	0.16	207.77	49.70
	(0.01)	(1.89)	(313.55)	(70.88)	(0.01)	(1.84)	(385.99)	(131.84)
T4 × Post	0.03***	6.51**	1299.60***	362.14***	0.02*	-0.19	443.44	194.84*
	(0.01)	(2.88)	(345.02)	(95.71)	(0.01)	(1.71)	(359.91)	(117.59)
Control mean	0.02	4.02	556.83	166.09	0.06	6.19	1,410.56	482.05
Control SD	0.15	40.63	4,690.96	1,366.70	0.24	39.22	6,718.40	2,319.12
Observations	164,802	164,802	164,802	164,802	164,802	164,802	164,802	164,802
Panel B: Weekend								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1 × Post	-0.00	1.94	-21.36	10.53	0.00	-0.32	53.57	-46.31
	(0.01)	(1.96)	(424.34)	(94.72)	(0.02)	(3.71)	(513.91)	(168.22)
T2 × Post	0.01	3.94	252.15	80.39	0.02	0.21	161.72	25.92
	(0.01)	(2.62)	(477.06)	(107.80)	(0.02)	(2.89)	(470.96)	(167.53)
T3 × Post	0.01	4.73*	318.63	133.90	0.04**	-0.43	786.62	218.03
	(0.01)	(2.46)	(447.12)	(104.41)	(0.02)	(2.92)	(535.51)	(182.96)
T4 × Post	0.03*	8.70**	1116.14**	324.28**	0.02	-2.96	466.73	174.82
	(0.01)	(3.58)	(521.55)	(132.41)	(0.02)	(3.08)	(627.86)	(208.85)
Control mean	0.02	2.73	484.12	142.29	0.07	5.99	1,840.03	555.37
Control SD	0.14	31.83	4,164.17	1,204.70	0.25	38.56	24,698.49	2,577.42
Observations	66,284	66,284	66,284	66,284	66,284	66,284	66,284	66,284

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Treatment Effects by Day of Week: Up to Two Months Post-Treatment

	Off-peak hours				Non-off-peak hours			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	=1 if any	min.	kWh	KRW	=1 if any	min.	kWh	KRW
Panel A: Weekday								
T1 × Post	0.01	2.11	538.34**	140.58*	0.01	0.20	115.41	10.39
	(0.01)	(2.05)	(265.98)	(78.81)	(0.01)	(1.70)	(292.50)	(99.84)
T2 × Post	0.02**	1.57	570.88**	150.30**	0.01	1.11	363.47	94.28
	(0.01)	(1.90)	(263.31)	(73.89)	(0.01)	(1.67)	(279.67)	(97.01)
T3 × Post	0.01	5.05	683.15*	182.51*	0.01	0.62	274.98	73.97
	(0.01)	(3.37)	(362.14)	(96.83)	(0.01)	(2.20)	(371.66)	(128.06)
T4 × Post	0.03***	9.65**	1677.42***	452.01***	0.02	-0.61	328.37	146.61
	(0.01)	(3.91)	(397.86)	(117.53)	(0.01)	(2.15)	(341.93)	(110.80)
Control mean	0.02	4.02	556.83	166.09	0.06	6.19	1,410.56	482.05
Control SD	0.15	40.63	4,690.96	1,366.70	0.24	39.22	6,718.40	2,319.12
Observations	173,882	173,882	173,882	173,882	173,882	173,882	173,882	173,882
Panel B: Weekend								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1 × Post	-0.00	1.74	-124.23	-22.12	0.01	1.99	371.23	84.08
	(0.01)	(2.95)	(377.03)	(109.68)	(0.01)	(2.94)	(405.91)	(132.50)
T2 × Post	0.01	1.03	241.46	37.01	0.02	1.37	323.91	85.06
	(0.01)	(2.63)	(531.95)	(122.05)	(0.01)	(2.22)	(369.83)	(129.14)
T3 × Post	0.02*	6.23*	701.54	224.88*	0.04***	1.33	773.72**	233.99*
	(0.01)	(3.55)	(450.83)	(128.26)	(0.01)	(2.42)	(392.55)	(134.92)
T4 × Post	0.03***	9.46**	1640.02***	463.29***	0.02	-0.71	643.86	251.17
	(0.01)	(3.72)	(527.96)	(151.58)	(0.02)	(2.14)	(474.68)	(156.12)
Control mean	0.02	2.73	484.12	142.29	0.07	5.99	1,840.03	555.37
Control SD	0.14	31.83	4,164.17	1,204.70	0.25	38.56	24,698.49	2,577.42
Observations	69,916	69,916	69,916	69,916	69,916	69,916	69,916	69,916

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Treatment Effects on Off-peak Charging Outcomes by Charger Type and Day of Week: Up to First Month Post-Treatment

	Fast chargers				Slow chargers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	=1 if any	min.	kWh	KRW	=1 if any	min.	kWh	KRW
Panel A: Weekday								
T1 × Post	0.01*	0.44	404.32*	101.07*	-0.00	1.16	143.64	46.37
	(0.01)	(0.27)	(231.96)	(57.99)	(0.01)	(1.99)	(175.08)	(57.58)
T2 × Post	0.02***	0.66**	496.28**	136.38***	0.01	2.10	327.72	101.75
	(0.01)	(0.27)	(217.63)	(50.55)	(0.01)	(2.04)	(219.36)	(72.08)
T3 × Post	0.01**	0.41	307.42	71.65	0.00	0.03	69.56	15.82
	(0.01)	(0.29)	(241.49)	(44.47)	(0.01)	(1.84)	(170.79)	(51.95)
T4 × Post	0.02***	0.70**	606.37**	138.95***	0.02*	5.81**	693.24***	223.19**
	(0.01)	(0.28)	(249.70)	(51.45)	(0.01)	(2.89)	(263.61)	(86.50)
Control mean	0.01	0.39	273.74	75.11	0.01	3.63	283.09	90.98
Control SD	0.11	4.41	3,431.75	888.84	0.11	40.41	3,211.23	1,041.07
Observations	164,802	164,802	164,802	164,802	164,802	164,802	164,802	164,802
Panel B: Weekend								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1 × Post	-0.00	-0.31	-192.30	-32.42	0.00	2.25	170.94	42.94
	(0.01)	(0.43)	(382.94)	(74.59)	(0.01)	(1.90)	(173.55)	(59.01)
T2 × Post	0.01	0.18	81.64	35.49	0.00	3.76	170.50	44.90
	(0.01)	(0.52)	(433.50)	(81.94)	(0.01)	(2.58)	(210.58)	(73.31)
T3 × Post	0.00	-0.07	-23.32	29.64	0.01	4.80**	341.94	104.26
	(0.01)	(0.42)	(381.62)	(75.04)	(0.01)	(2.40)	(216.07)	(71.21)
T4 × Post	0.02	0.48	350.11	93.88	0.01	8.23**	766.03**	230.39**
	(0.01)	(0.50)	(417.64)	(81.25)	(0.01)	(3.57)	(333.51)	(108.97)
Control mean	0.01	0.38	280.95	75.90	0.01	2.34	203.16	66.39
Control SD	0.10	4.12	3,203.32	835.24	0.10	31.54	2,655.68	865.13
Observations	66,284	66,284	66,284	66,284	66,284	66,284	66,284	66,284

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Treatment Effects on Off-peak Charging Outcomes by Charger Type and Day of Week: Up to Two Months Post-Treatment

	Fast chargers				Slow chargers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	=1 if any	min.	kWh	KRW	=1 if any	min.	kWh	KRW
Panel A: Weekday								
T1 × Post	0.01**	0.44*	394.58**	93.61**	-0.00	1.67	143.75	46.96
	(0.01)	(0.23)	(183.70)	(44.32)	(0.00)	(2.03)	(190.83)	(66.26)
T2 × Post	0.02***	0.63**	453.19**	119.11***	0.00	0.94	117.69	31.19
	(0.01)	(0.26)	(202.50)	(43.65)	(0.01)	(1.90)	(185.92)	(62.92)
T3 × Post	0.01**	0.51	392.51	88.49*	0.00	4.54	290.65	94.03
	(0.00)	(0.32)	(248.01)	(45.32)	(0.01)	(3.35)	(264.55)	(87.06)
T4 × Post	0.02***	1.00***	836.85***	180.64***	0.01	8.65**	840.57**	271.36**
	(0.01)	(0.30)	(260.61)	(49.47)	(0.01)	(3.93)	(331.28)	(113.03)
Control mean	0.01	0.39	273.74	75.11	0.01	3.63	283.09	90.98
Control SD	0.11	4.41	3,431.75	888.84	0.11	40.41	3,211.23	1,041.07
Observations	173,882	173,882	173,882	173,882	173,882	173,882	173,882	173,882
Panel B: Weekend								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1 × Post	-0.00	-0.33	-232.62	-42.61	0.00	2.07	108.39	20.49
	(0.01)	(0.31)	(262.93)	(53.06)	(0.01)	(2.93)	(276.98)	(98.05)
T2 × Post	0.01	0.41	343.33	84.72	0.00	0.63	-101.87	-47.71
	(0.01)	(0.60)	(483.34)	(88.51)	(0.01)	(2.57)	(237.53)	(87.49)
T3 × Post	0.01	0.29	275.70	85.99	0.01	5.94*	425.84	138.89
	(0.01)	(0.36)	(322.49)	(64.76)	(0.01)	(3.53)	(314.61)	(111.47)
T4 × Post	0.02**	0.95*	774.54*	183.15**	0.01	8.51**	865.48**	280.14**
	(0.01)	(0.49)	(395.45)	(78.57)	(0.01)	(3.72)	(376.88)	(135.73)
Control mean	0.01	0.38	280.95	75.90	0.01	2.34	203.16	66.39
Control SD	0.10	4.12	3,203.32	835.24	0.10	31.54	2,655.68	865.13
Observations	69,916	69,916	69,916	69,916	69,916	69,916	69,916	69,916

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Elasticity Calculation

Method. The discount in each arm k is applied as a percentage off the standard off-peak rate, so the effective off-peak price faced by a participant, expressed as a fraction of the standard rate p^{std}, equals

$$p_k \;=\; (1 - d_k)\, p^{\text{std}},$$

where $d_k \in \{0,\, 0.20,\, 0.40\}$ denotes the discount fraction for the control, 20%-off, and 40%-off arms respectively. Comparing the 40%-off arms (T2, T4) against the 20%-off arms (T1, T3) holds the "treated-and-aware" status constant and isolates the price margin: $p_{40\%} = 0.60\, p^{\text{std}}$ and $p_{20\%} = 0.80\, p^{\text{std}}$. The midpoint (arc) price reduction between these two arms is therefore independent of the underlying KRW value of p^{std}:

$$\frac{\Delta p}{\bar{p}} \;=\; \frac{p_{40\%} - p_{20\%}}{(p_{40\%} + p_{20\%})/2} \;=\; \frac{0.60 - 0.80}{0.70} \;=\; -28.57\%.$$

For each arm, the implied per-user-day off-peak energy under treatment is the pre-treatment control mean M^{off} plus the regression coefficient β_k^{off} from Table 3,

$$Q_k \;=\; M^{\text{off}} + \beta_k^{\text{off}}.$$

The arc price elasticity of off-peak energy is then

$$\sigma \;=\; \frac{\Delta Q/\bar{Q}}{\Delta p/\bar{p}} \;=\; \frac{(Q_{40\%} - Q_{20\%})/\big[(Q_{40\%} + Q_{20\%})/2\big]}{(p_{40\%} - p_{20\%})/\big[(p_{40\%} + p_{20\%})/2\big]}. \tag{3}$$

Computation. We compute equation (3) in three forms: a pooled estimate that combines the two 40%-off arms (T2, T4) and the two 20%-off arms (T1, T3) over the two-month observation window, and two within-duration estimates that hold exposure length constant. For the pooled estimate, using the off-peak energy coefficients from Table 3, Panel B (in the unit of kWh per user-day; see footnote on unit convention),

$$\begin{aligned}
\beta_{40\%}^{\text{off}} \;&=\; \tfrac{1}{2}\,(\beta_{T2}^{\text{off}} + \beta_{T4}^{\text{off}}) \;=\; \tfrac{1}{2}\,(0.474 + 1.657) \;=\; 1.066,\\
\beta_{20\%}^{\text{off}} \;&=\; \tfrac{1}{2}\,(\beta_{T1}^{\text{off}} + \beta_{T3}^{\text{off}}) \;=\; \tfrac{1}{2}\,(0.346 + 0.686) \;=\; 0.516,
\end{aligned}$$

so that $Q_{40\%} = 0.536 + 1.066 = 1.602$ and $Q_{20\%} = 0.536 + 0.516 = 1.052$, giving $\Delta Q = 0.549$ and $\bar{Q} = 1.327$. The percent change in quantity is $\Delta Q/\bar{Q} = 41.41\%$, and substituting into equation (3),

$$\sigma_{\text{pooled}} \;=\; \frac{+41.41\%}{-28.57\%} \;\approx\; -1.45.$$

The within-duration estimates follow the same algebra. For the one-month arms, using $\beta_{T1}^{\text{off}} = 0.384$ and $\beta_{T2}^{\text{off}} = 0.659$ (Table 3, Panel A), we obtain $\Delta Q/\bar{Q} = 26.04\%$ and $\sigma_{1\,\text{mo}} \approx -0.91$. For the two-month arms, using $\beta_{T3}^{\text{off}} = 0.686$ and $\beta_{T4}^{\text{off}} = 1.657$ (Table 3, Panel B), we obtain $\Delta Q/\bar{Q} = 56.88\%$ and $\sigma_{2\,\text{mo}} \approx -1.99$.

Survey Overview

Survey Period

- August 1 (Fri) – August 17 (Sun), 2025

Purpose of the Survey and Research

Joint study by Elecvery × KDI School of Public Policy and Management

- Recruiting participants for the study: "Research on Time-of-Use EV Charging Discounts"
- Collecting data to support policy research on future EV adoption and improving the utilization of charging infrastructure

Survey Content

- Basic Elecvery member information, basic demographics, EV and charging experience, etc. (and gift selection!)
- Estimated time: about 3–4 minutes

Participation Rewards

Within 3 days after the survey period ends, we will send you one mobile coupon of your choice.

Choose 1 from the options!

- Coupons cannot be reissued for any reason.

Important Notes

If fraudulent responses are detected, or if your Elecvery membership information cannot be verified: your submitted survey will be invalidated without notice, and you will be excluded from the participation rewards and the discount program.

Eligibility Check

This survey is intended only for those who meet **all** of the following conditions:

- Have owned an electric vehicle for 3 months or more
- Have charged for at least KRW 5,000 using an Elecvery membership within the last 30 days
- Have agreed to provide three types of member identification information: name, mobile phone number, and primary account information
- Are able to respond sincerely to all survey questions

  [label=○]

- I meet the criteria. (Continue)
- I do not meet the criteria. (End survey)

Consent to Collection and Use of Personal Information

TBU Co., Ltd. intends to collect, use, and provide the following personal information to third parties for the stated purposes. Please read the details carefully before deciding whether to consent.

Collection and Use of Personal Information

Items	**Purpose of Collection and Use**	**Retention Period**
Name, contact (mobile phone number), login method, and survey responses	Selection of event participants, reward provision, analysis and improvement of service use, survey purposes, etc.	Collected information is stored for up to 6 months after the service ends, then automatically destroyed.

Provision of Personal Information to Third Parties

tableheader **Recipient**	**Purpose**	**Items Provided**	**Retention Period**
KDI School of Public Policy and Management	Service provision, selection of target participants, reward provision, analysis and improvement of service use, survey, and research on EV charging use	Survey responses **excluding** name, contact (mobile phone number), and login method; charging volume, charging cost, charging time	Collected information is stored for up to 6 months after the service ends, then automatically destroyed.

∗ Respondents have the right to refuse consent to the collection and use of the above personal information.

If you refuse consent, your use of services requiring personal information may be limited.

Do you consent to the collection and use of personal information as described above?

[label=○]

- I consent
- I do not consent (cannot participate in the event; survey ends)

Research Participation Consent Form

Hello,

This is an online survey conducted as part of a research study designed to understand the charging behavior of EV users. If you respond to this survey, you may subsequently participate in a research study that includes experimental interventions. This research is conducted by the KDI School of Public Policy and Management, and the results may help improve public policy and EV infrastructure in the future.

1. Research Purpose

This research aims to experimentally examine various factors (e.g., information provision, small incentives) that may influence the charging behavior of EV users.

2. Research Content and Procedures

- First, you will participate in this survey (baseline survey, approximately 4 minutes).
- Afterward, over a certain period, you will be randomly assigned to various conditions that may influence your EV charging behavior. Some participants may receive information about charging behavior or small rewards, while others may receive no additional information or rewards.
- During the research period, your charging records (charging date/time, amount, cost, etc.) will be collected through the Elecvery app and TBU Co., Ltd., and all data will be provided to the research team in anonymized form. Your personal information (e.g., name, phone number) will **not** be transferred to the research team, and the data will be used solely for research purposes.

3. Privacy Protection and Data Use

- All collected information will be anonymized and used only for research purposes. It will not be provided to any third parties.
- All data will be managed according to strict security procedures.

4. Voluntary Participation and Right to Withdraw

- Participation in this research is entirely voluntary, and you may stop participating at any time without reason.
- There will be no disadvantage to you for discontinuing participation.

5. Contact

If you have any questions or concerns regarding this research, please feel free to contact us:

- Principal Investigator: Professor Young Jae Kim (KDI School of Public Policy and Management)
- Email: kyj@kdischool.ac.kr

Having fully understood the above, do you voluntarily agree to participate in this survey and the subsequent research?

[label=○]

- I agree and will participate in the research.

- I do not agree and will not participate in the research.

Survey Questions

Q1. Member Identification – Your Name (1/20)

Please enter the name you registered with Elecvery.
Example: Hong Gildong
∗ If your information cannot be found in the member database, you will be excluded from this event.

Q2. Member Identification – Mobile Phone Number (2/20)

Please enter the mobile phone number you registered with Elecvery.
Example: 010-1234-5678
∗ If your information cannot be found in the member database, you will be excluded from this event.

Q3. Member Identification – Primary Account (3/20)

Please select your primary Elecvery account.
Example: Kakao
∗ If your information cannot be found in the member database, you will be excluded from this event.
∗ You can check this under **Settings** > **Manage My Info** > **My elecvery Account**.

- Google
- Apple
- Naver
- Kakao
- Vehicle account (Tesla, Hyundai, Kia, Genesis)

Q4. Please tell us your gender. (4/20)

- Male
- Female

Q5. Please select your age group. (5/20)

- Under 20
- 20s
- 30s
- 40s
- 50s
- 60 or older

Q6. Highest level of education (based on graduation). (6/20)

- High school or below
- Associate degree (2-year college)
- Bachelor's degree
- Master's degree
- Doctoral degree or above

Q7. Current region of residence. (7/20)

Please select a province/metropolitan city:

- Seoul, Busan, Daegu, Incheon, Gwangju, Daejeon, Ulsan, Sejong
- Gyeonggi-do, Gangwon Special Self-Governing Province
- Chungcheongbuk-do, Chungcheongnam-do
- Jeollabuk-do, Jeollanam-do
- Gyeongsangbuk-do, Gyeongsangnam-do
- Jeju Special Self-Governing Province

Please enter your city/county/district (e.g., Songpa-gu, Sangju-si, Seogwipo-si).

Q8. Area where you primarily work. (8/20)

Please select a province/metropolitan city:

- Seoul, Busan, Daegu, Incheon, Gwangju, Daejeon, Ulsan, Sejong
- Gyeonggi-do, Gangwon Special Self-Governing Province
- Chungcheongbuk-do, Chungcheongnam-do
- Jeollabuk-do, Jeollanam-do
- Gyeongsangbuk-do, Gyeongsangnam-do
- Jeju Special Self-Governing Province

Please enter your city/county/district (e.g., Songpa-gu, Sangju-si, Seogwipo-si).

Q9. Current type of housing. (9/20)

If your type is not listed, please choose the closest option.

- Apartment
- Officetel (studio/mixed-use apartment)
- Detached house
- Villa (low-rise multi-family housing)

Q10. Approximate annual income. (10/20)

- Less than KRW 30 million
- KRW 30 million – 50 million
- KRW 50 million – 70 million
- KRW 70 million – 100 million
- KRW 100 million or more

Q11. Employment type closest to yours. (11/20)

- Full-time (e.g., 9 a.m. – 6 p.m.)
- Part-time
- Flexible work arrangement
- Remote / work-from-home
- Self-employed
- Student / homemaker
- Unemployed

Q12. Which electric vehicle do you own? (12/20)

If you own one, please enter the [**manufacturer, model, year, range**].
Examples: Kia EV6 2022 Standard; Tesla Model 3 2019 LR (Long Range), etc.

Q13. How long have you owned your electric vehicle? (13/20)

- Less than 3 months
- 3 – 6 months
- 6 months – 1 year
- 1 – 3 years
- 3 years or more

Q14. On average, how many times do you charge per month? (14/20)

Include charging sessions using cards other than Elecvery.
Example: Elecvery card 3 times + Ministry of Environment card 2 times = 5 times

- 1 time
- 2 times
- 3 times

- 4 times
- 5 times
- 6–7 times
- 8–9 times
- 10 or more times

Q15. Approximate annual mileage (km). (15/20)

Tip: Please select the option closest to the annual mileage you reported to your insurance company (or last year's mileage).

- Less than 3,000 km
- 3,000 – 5,000 km
- 5,000 – 7,000 km
- 7,000 – 10,000 km
- 10,000 – 15,000 km
- 15,000 km or more

Q16. Average monthly charging cost. (16/20)

Include charging sessions using cards other than Elecvery.
(Example: Elecvery approx. KRW 20,000 + Ministry of Environment approx. KRW 15,000 = select KRW 30,000 – 50,000)

- Less than KRW 10,000
- KRW 10,000 – 20,000
- KRW 20,000 – 30,000
- KRW 30,000 – 50,000
- KRW 50,000 – 70,000
- KRW 70,000 – 100,000
- KRW 100,000 or more

Q17. Chargers installed where you currently live. (17/20)

Based on within a 10-minute walk from home.

- Both slow and fast chargers available
- Only slow chargers available
- Only fast chargers available
- No chargers within a 10-minute walk / don't know

Q18. Chargers installed at your current workplace. (18/20)

Based on within a 10-minute walk from the workplace.

- Both slow and fast chargers available
- Only slow chargers available
- Only fast chargers available
- No chargers within a 10-minute walk / don't know

Q19. Where do you usually charge? (19/20)

In general, regardless of whether it's through Elecvery!

- Home
- Workplace
- Highway (rest areas, drowsy-driving rest spots)
- Public facilities (government offices, parks, supermarkets, etc.)

Q20. What time of day do you usually charge? (20/20)

In general, regardless of whether it's through Elecvery!

- Morning (06:00–12:00)
- Afternoon (12:00–18:00)
- Evening (18:00–22:00)

- Night (22:00–06:00)

Thank you for participating in the survey!